Statistics

Chapter 6 — Instructor Solutions Manual

October 10, 2026 • Total Exercises: 71

Chapter 6 **Exercise 1**

Page 193

PROBLEM STATEMENT

Chapter 6, Page 193

Step 1

The formula for the mean is

$$\mu = \Sigma[xP(x)]$$
$$\mu = 0 \times 0.2 + 1 \times 0.4 + 2 \times 0.3 + 3 \times 0.1$$
$$\mu = 0.4 + 0.6 + 0.3$$
$$\mu = 1.3$$

The formula for the variance is

$$\sigma^2 = \Sigma[(x-\mu)^2 P(x)]$$
$$\sigma^2 = (0-1.3)^2 \times 0.2 + (1-1.3)^2 \times 0.4 + (2-1.3)^2 \times 0.3 + (3-1.3)^2 \times 0.1$$
$$\sigma^2 = (-1.3)^2 \times 0.2 + (0.3)^2 \times 0.4 + (0.7)^2 \times 0.3 + (1.7)^2 \times 0.1$$
$$\sigma^2 = 1.69 \times 0.2 + 0.09 \times 0.4 + 0.49 \times 0.3 + 2.89 \times 0.1$$
$$\sigma^2 = 0.338 + 0.036 + 0.147 + 0.289$$
$$\sigma^2 = 0.81$$

Result

$\mu = 1.3$ and $\sigma^2 = 0.81$

PROBLEM STATEMENT

Chapter 6, Page 193

Step 1

The formula for the mean is

$$\mu = \Sigma[xP(x)]$$
$$\mu = 2 \times 0.5 + 8 \times 0.3 + 10 \times 0.2$$
$$\mu = 1 + 2.4 + 2$$
$$\mu = 5.4$$

The formula for the variance is

$$\sigma^2 = \Sigma[(x-\mu)^2 P(x)]$$
$$\sigma^2 = (2-5.4)^2 \times 0.5 + (8-5.4)^2 \times 0.3 + (10-5.4)^2 \times 0.2$$
$$\sigma^2 = 11.56 \times 0.5 + 6.76 \times 0.3 + 21.16 \times 0.2$$
$$\sigma^2 = 5.78 + 2.028 + 4.232$$
$$\sigma^2 = 12.04$$

Result

$\mu = 5.4$ and $\sigma^2 = 12.04$

PROBLEM STATEMENT

Chapter 6, Page 194

Step 1

The formula for the mean is

$$\mu = \Sigma[xP(x)]$$
$$\mu = 5 \times 0.1 + 10 \times 0.3 + 15 \times 0.2 + 20 \times 0.4$$
$$\mu = 0.5 + 3 + 3 + 8$$
$$\mu = 14.5$$

The formula for the variance is

$$\sigma^2 = \Sigma[(x-\mu)^2 P(x)]$$
$$\sigma^2 = (5-14.5)^2 \times 0.1 + (10-14.5)^2 \times 0.3 + (15-14.5)^2 \times 0.2 + (20-14.5)^2 \times 0.4$$
$$\sigma^2 = 90.25 \times 0.1 + 20.25 \times 0.3 + 0.25 \times 0.2 + 30.25 \times 0.4$$
$$\sigma^2 = 9.025 + 6.075 + 0.05 + 12.1$$
$$\sigma^2 = 27.25$$

Result

$\mu = 14.5$ and $\sigma^2 = 27.25$

PROBLEM STATEMENT

Chapter 6, Page 194

Step 1

Discrete data are restricted to defined separated values, for example integers or counts. **Continuous** data are not restricted to defined separate values, but can occupy any value over a continuous range, for example decimal, rational or real numbers.

(a) Discrete, because the number of new accounts are counts (that is, nonnegative integers).

(b) Continuous, because the time between customer arrivals can take on decimal values (such as 2.5 minutes or 23.75 seconds).

(c) Discrete, because the number of customers are counts (that is, nonnegative integers).

(d) Continuous, because the amount of fuel can take on decimal values (such as 14.5 gallons).

(e) Discrete, because the number of minorities are counts (that is, nonnegative integers).

(f) Continuous, because the outside temperature can take on decimal values (such as 7.5° C).

Result

(a) Discrete

(b) Continuous

(c) Discrete

(d) Continuous

(e) Discrete

(f) Continuous

PROBLEM STATEMENT

Chapter 6, Page 194

Step 1

Part (a): To get the probability distribution, divide the frequency of number of calls with the total number of frequencies or last 50 days

$$\begin{array}{ll} \text{Number of calls } x & \text{Pobability of call } P(x) \\ 0 & \frac{8}{50} = 0.16 \\ 1 & \frac{10}{50} = 0.5 \\ 2 & \frac{22}{50} = 0.44 \\ 3 & \frac{9}{50} = 0.18 \\ 4 & \frac{1}{50} = 0.02 \end{array}$$

Part (b): This is an example of a discrete distribution.

Step 2

Part (c): The mean number of emergency calls per day is

$$\begin{aligned} \mu &= \Sigma[xP(x)] \\ \mu &= 0 \times 0.16 + 1 \times 0.5 + 2 \times 0.44 + 3 \times 0.18 + 4 \times 0.02 \\ \mu &= 0 + 0.5 + 0.88 + 0.54 + 0.08 \\ \mu &= 2 \end{aligned}$$

Part (d): The variance of emergency calls per day is

$$\begin{aligned} \sigma^2 &= \Sigma[(x-\mu)^2 P(x)] \\ \sigma^2 &= (0-2)^2 \times 0.16 + (1-2)^2 \times 0.5 + (2-2)^2 \times 0.44 + (3-2)^2 \times 0.18 + (4-2)^2 \times 0.02 \\ \sigma^2 &= 4 \times 0.16 + 1 \times 0.5 + 0 \times 0.44 + 1 \times 0.18 + 4 \times 0.02 \\ \sigma^2 &= 0.64 + 0.5 + 0 + 0.18 + 0.08 \\ \sigma^2 &= 1.4 \end{aligned}$$

The standard deviation is

$$\begin{aligned} \sigma &= \sqrt{\sigma^2} \\ \sigma &= \sqrt{1.4} \\ \sigma &= 1.18 \end{aligned}$$

Result

Part (a): Done

Part (b): Discrete Distribution

Part (c): $\mu = 2$

Part (d): $\sigma =$

PROBLEM STATEMENT

Chapter 6, Page 194

Step 1

Expected number of admissions for the fall semester is

$$\begin{aligned}\mu &= \Sigma[xP(x)]\\ \mu &= 1000 \times 0.6 + 1200 \times 0.3 + 1500 \times 0.1\\ \mu &= 600 + 360 + 150\\ \mu &= 1110\end{aligned}$$

Variance of the number of admissions is

$$\begin{aligned}\sigma^2 &= \Sigma[(x-\mu)^2 P(x)]\\ \sigma^2 &= (1000-1110)^2 \times 0.6 + (1200-1110)^2 \times 0.3 + (1500-1110)^2 \times 0.1\\ \sigma^2 &= 12100 \times 0.6 + 8100 \times 0.3 + 152100 \times 0.1\\ \sigma^2 &= 7260 + 2430 + 15210\\ \sigma^2 &= 24900\end{aligned}$$

Standard deviation is

$$\begin{aligned}\sigma &= \sqrt{\sigma^2}\\ \sigma &= \sqrt{24900}\\ \sigma &= 157.8\end{aligned}$$

Result

expected number of admissions is 1110, Variance is 24900 and standard deviation is 157.8

PROBLEM STATEMENT

Chapter 6, Page 194

Step 1

Part (a): The mean amount deducted from the total purchase account is

$$\begin{aligned}\mu &= \Sigma[xP(x)]\\ \mu &= 10\times 0.5+25\times 0.4+50\times 0.08+100\times 0.02\\ \mu &= 5+10+4+2\\ \mu &= 21\end{aligned}$$

The mean amount deducted from the total purchase account is $\$21$

Part (b): The variance of the amount deducted from the total purchase account is

$$\begin{aligned}\sigma^2 &= \Sigma[(x-\mu)^2P(x)]\\ \sigma^2 &= (10-21)^2\times 0.5+(25-21)^2\times 0.4+(50-21)^2\times 0.08+(100-21)^2\times 0.02\\ \sigma^2 &= 121\times 0.5+16\times 0.4+841\times 0.08+3241\times 0.02\\ \sigma^2 &= 60.5+6.4+67.28+64.82\\ \sigma^2 &= 199\end{aligned}$$

The standard deviation is

$$\begin{aligned}\sigma &= \sqrt{\sigma^2}\\ \sigma &= \sqrt{199}\\ \sigma &= 14.107\end{aligned}$$

The standard deviation of the amount deducted from the total purchase account is $\$14.107$

Result

$\mu = 21$ and $\sigma = 14.107$

PROBLEM STATEMENT

Chapter 6, Page 195

Step 1

a. To get the probability distribution, divide the frequency of parking for each hour with the total number of frequencies or sample consumers of 250

Number of hours x	Pobability of parking $P(x)$	Amount Charged y
1	$\frac{20}{250} = 0.08$	3
2	$\frac{38}{250} = 0.152$	6
3	$\frac{53}{250} = 0.212$	9
4	$\frac{45}{250} = 0.18$	12
5	$\frac{40}{250} = 0.16$	14
6	$\frac{13}{250} = 0.052$	16
7	$\frac{5}{250} = 0.02$	18
8	$\frac{36}{250} = 0.08$	20

Step 2

b. The mean of the number of hours parked μ_x

$$\begin{aligned}\mu_x &= \Sigma[xP(x)]\\ \mu_x &= 1\times 0.08 + 2\times 0.152 + 3\times 0.212 + 4\times 0.18 + 5\times 0.16 + 6\times 0.052 + 7\times 0.02 + 8\times 0.08\\ \mu_x &= 0.08 + 0.304 + 0.636 + 0.72 + 0.8 + 0.312 + 0.14 + 0.64\\ \mu_x &= 3.632\end{aligned}$$

The variance of the number of hours parked is

$$\begin{aligned}\sigma_x^2 =&\Sigma[(x-\mu_x)^2 P(x)]\\ \sigma_x^2 =&(1-3.632)^2\times 0.08 + (2-3.632)^2\times 0.152 + (3-3.632)^2\times 0.212 + (4-3.632)^2\times 0.18\\ &+ (5-3.632)^2\times 0.16 + (6-3.632)^2\times 0.052 + (7-3.632)^2\times 0.02 + (8-3.632)^2\times 0.08\\ \sigma_x^2 =& 6.93\times 0.08 + 2.66\times 0.152 + 0.4\times 0.212 + 0.135\times 0.18 + 1.87\times 0.16 + 5.61\times 0.052\\ &+ 11.34\times 0.02 + 19.08\times 0.08\\ \sigma_x^2 =& 0.5544 + 0.40432 + 0.0848 + 0.0243 + 0.2992 + 0.29172 + 0.2268 + 19.16\\ \sigma_x^2 =& 20.81874\end{aligned}$$

The standard deviation is

$$\begin{aligned}\sigma_x &= \sqrt{\sigma_x^2}\\ \sigma_x &= \sqrt{20.81874}\\ \sigma_x &= 4.56\end{aligned}$$

A typical consumer parks for about 3.632 hours on an average.

Step 3

c. The mean of the amount charged μ_y

$$\begin{aligned}
\mu_y &= \Sigma[yP(x)] \\
\mu_y &= 3 \times 0.08 + 6 \times 0.152 + 9 \times 0.212 + 12 \times 0.18 + 14 \times 0.16 + 16 \times 0.052 + 18 \times 0.02 + 20 \times 0.08 \\
\mu_y &= 0.24 + 0.912 + 1.908 + 2.16 + 2.24 + 0.832 + 0.36 + 1.6 \\
\mu_y &= 10.252
\end{aligned}$$

The variance of the amount charged is

$$\begin{aligned}
\sigma_y^2 =& \Sigma[(y-\mu_y)^2 P(x)] \\
\sigma_y^2 =& (3-10.252)^2 \times 0.08 + (6-10.252)^2 \times 0.152 + (9-10.252)^2 \times 0.212 \\
&+ (12-10.252)^2 \times 0.18 + (14-10.252)^2 \times 0.16 + (16-10.252)^2 \times 0.052 \\
&+ (18-10.252)^2 \times 0.02 + (20-10.252)^2 \times 0.08 \\
\sigma_y^2 =& 52.6 \times 0.08 + 18.08 \times 0.152 + 1.57 \times 0.212 + 3.05 \times 0.18 + 14.05 \times 0.16 \\
&+ 33.04 \times 0.052 + 60.03 \times 0.02 + 95.02 \times 0.08 \\
\sigma_y^2 =& 4.208 + 2.75 + 0.333 + 0.549 + 2.248 + 1.72 + 1.2006 + 7.60 \\
\sigma_y^2 =& 20.6086
\end{aligned}$$

The standard deviation is

$$\begin{aligned}
\sigma_y &= \sqrt{\sigma_y^2} \\
\sigma_y &= \sqrt{20.6086} \\
\sigma_y &= 4.54
\end{aligned}$$

Result

Part (a): Done

Part (b): $\mu_x = 3.632$ and $\sigma_x = 4.56$

Part (c): $\mu_y = 10.252$ and $\sigma_y = 4.54$

PROBLEM STATEMENT

Chapter 6, Page 201

Step 1

$n = 4$ and $\pi = 0.25$

Part (a): $x = 2$

$$\begin{aligned} P(x = 2) &= {}_n C_x \pi^x (1 - \pi)^{n-x} \\ &= {}_4 C_2 (0.25)^2 (1 - 0.25)^{4-2} \\ &= 3 \times (0.0625) \times (0.75)^2 \\ &= 0.125 \times (0.5625) \\ &= 0.0703125 \end{aligned}$$

Part (b): $x = 3$

$$\begin{aligned} P(x = 3) &= {}_n C_x \pi^x (1 - \pi)^{n-x} \\ &= {}_4 C_3 (0.25)^3 (1 - 0.25)^{4-3} \\ &= 4 \times (0.015625) \times (0.75) \\ &= 0.0625 \times (0.75) \\ &= 0.046875 \end{aligned}$$

Result

$P(x = 2) = 0.0703125$ and $P(x = 3) = 0.046875$

PROBLEM STATEMENT

Chapter 6, Page 201

Step 1

$n = 5$ and $\pi = 0.40$

Part (a): $x = 1$

$$\begin{aligned} P(x=1) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_5 C_1 (0.40)^1 (1-0.40)^{4-1} \\ &= 5 \times (0.40) \times (0.60)^3 \\ &= 2 \times (0.216) \\ &= 0.432 \end{aligned}$$

Part (b): $x = 2$

$$\begin{aligned} P(x=2) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_5 C_2 (0.40)^2 (1-0.40)^{4-2} \\ &= 10 \times (0.16) \times (0.60)^2 \\ &= 1.60 \times (0.36) \\ &= 0.576 \end{aligned}$$

Result

$P(x=1) = 0.432$ and $P(x=2) = 0.576$

PROBLEM STATEMENT

Chapter 6, Page 201

Step 1

Given:

$$n = \text{Sample size} = 3$$
$$\pi = \text{Probability of success} = 0.60$$

(a) The probability $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, and $P(X = 3)$ are given in the column starting with $\pi = 0.60$ of the Binomial probability distribution table with $n = 3$ in the appendix. The probabilities $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, and $P(X = 3)$ are given in the order from top to bottom.

$$P(X = 0) = 0.064$$
$$P(X = 1) = 0.288$$
$$P(X = 2) = 0.432$$
$$P(X = 3) = 0.216$$

Step 2

(b) The mean of a binomial distribution is the product of the sample size n and the probability π:

$$\mu = n\pi = 3 \times 0.6 = 1.8$$

The standard deviation of a binomial distribution is the square root of the product of the sample size n and the probabilities π and $1 - \pi$ (that is, the standard deviation is the square root of the variance).

$$\sigma = \sqrt{n\pi(1-\pi)} = \sqrt{3(0.6)(1-0.6)} \approx 0.8485$$

Result

(a)

$$P(X = 0) = 0.064$$
$$P(X = 1) = 0.288$$
$$P(X = 2) = 0.432$$
$$P(X = 3) = 0.216$$

(b) Mean 1.8 and Standard deviation 0.8485

PROBLEM STATEMENT

Chapter 6, Page 201

Step 1

Given:

$$n = \text{Sample size} = 5$$
$$\pi = \text{Probability of success} = 0.30$$

(a) The probability $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, $P(X = 3)$, $P(X = 4)$, and $P(X = 5)$ are given in the column starting with $\pi = 0.30$ of the Binomial probability distribution table with $n = 5$ in the appendix. The probabilities $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, $P(X = 3)$, $P(X = 4)$, and $P(X = 5)$ are given in the order from top to bottom.

$$P(X = 0) = 0.168$$
$$P(X = 1) = 0.360$$
$$P(X = 2) = 0.309$$
$$P(X = 3) = 0.132$$
$$P(X = 4) = 0.028$$
$$P(X = 5) = 0.002$$

Step 2

(b) The mean of a binomial distribution is the product of the sample size n and the probability π:

$$\mu = n\pi = 5 \times 0.30 = 1.5$$

The standard deviation of a binomial distribution is the square root of the product of the sample size n and the probabilities π and $1 - \pi$ (that is, the standard deviation is the square root of the variance).

$$\sigma = \sqrt{n\pi(1 - \pi)} = \sqrt{5(0.30)(1 - 0.30)} \approx 1.0247$$

Result

(a)

$$P(X = 0) = 0.168$$
$$P(X = 1) = 0.360$$
$$P(X = 2) = 0.309$$
$$P(X = 3) = 0.132$$
$$P(X = 4) = 0.028$$
$$P(X = 5) = 0.002$$

(b) Mean 1.5 and Standard deviation 1.0247

PROBLEM STATEMENT

Chapter 6, Page 201

Step 1

The probability that individual investor has used a discount broker (π)= 30% or 0.3

Total individuals (n)= 9

Part (a): Probability that two of the sampled individuals have used a discount broker $(P(x=2))$

$$\begin{aligned} P(x=2) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_9 C_2 (0.30)^2 (1-0.30)^{9-2} \\ &= 36 \times (0.09) \times (0.70)^7 \\ &= 3.24 \times 0.0403536 \\ &= 0.1307 \end{aligned}$$

Part (b): Probability that four of the sampled individuals have used a discount broker $(P(x=4))$

$$\begin{aligned} P(x=4) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_9 C_4 (0.30)^4 (1-0.30)^{9-4} \\ &= 0.172 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (c): Probability that none of them have used a discount broker $(P(x=0))$

$$\begin{aligned} P(x=0) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_9 C_0 (0.30)^0 (1-0.30)^{9-0} \\ &= 0.040 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Result

Part (a): 0.1307

Part (b): 0.172

Part (c): 0.040

PROBLEM STATEMENT

Chapter 6, Page 201

Step 1

The probability that the first class mail within the same city is delivered within two days of the time of mailing (π)= 95% or 0.95

Number of observations (n)= 6

Part (a): Probability that all six arrive within two days $(P(x=6))$

$$\begin{aligned} P(x=6) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_6 C_6 (0.95)^6 (1-0.95)^{6-6} \\ &= 0.735 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (b): Probability that exactly five arrive within two days $(P(x=5))$

$$\begin{aligned} P(x=5) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_6 C_5 (0.95)^5 (1-0.95)^{6-5} \\ &= 0.232 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (c): The mean number of letters that will arrive within two days

$$\begin{aligned} \mu &= n\pi \\ &= 6 \times 0.95 \\ &= 5.7 \end{aligned}$$

Part (d): The variance that number of letters that will arrive within two days

$$\begin{aligned} \sigma^2 &= n\pi(1-\pi) \\ &= 6 \times 0.95 \times (1-0.5) \\ &= 0.285 \end{aligned}$$

The standard deviation is

$$\begin{aligned} \sigma &= \sqrt{\sigma^2 2} \\ &= \sqrt{0.285} \\ &= 0.534 \end{aligned}$$

Result

Part (a): 0.735

Part (b): 0.232

Part (c): 5.7

Part (d): $\sigma^2 = 0.285$ and $\sigma = 0.534$

PROBLEM STATEMENT

Chapter 6, Page 201

Step 1

The probability that new vehicles require warranty service within the first year is (π)= 10% and 0.1

Total cars sold (n)= 12

Part (a): Probability that none of these vehicles requires warranty service $(P(x=0))$

$$\begin{aligned} P(x=0) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{12} C_0 (0.10)^0 (1-0.10)^{12-0} \\ &= 0.282 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (b): Probability that exactly one of these vehicles requires warranty service $(P(x=1))$

$$\begin{aligned} P(x=1) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{12} C_1 (0.10)^1 (1-0.10)^{12-1} \\ &= 0.377 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (c): Probability that exactly two of these vehicles require warranty service $(P(x=2))$

$$\begin{aligned} P(x=2) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{12} C_2 (0.10)^2 (1-0.10)^{12-2} \\ &= 0.230 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (d): Mean of this probability distribution

$$\begin{aligned} \mu &= n\pi \\ &= 12 \times 0.1 \\ &= 1.2 \end{aligned}$$

Variance of this probability distribution

$$\begin{aligned} \sigma^2 &= n\pi(1-\pi) \\ &= 12 \times 0.1 \times (1-0.1) \\ &= 1.08 \end{aligned}$$

Step 2

Standard deviation is

$$\begin{aligned} \sigma &= \sqrt{\sigma^2} \\ &= \sqrt{1.08} \\ &= 1.04 \end{aligned}$$

Result

Part (a): 0.282

Part (b): 0.377

Part (c): 0.230

Part (d): $\mu = 1.2$ and $\sigma = 1.04$

PROBLEM STATEMENT

Chapter 6, Page 202

Step 1

Calls made in one hour =6

So, Calls made in two hours (n)=$6 \times 2 = 12$

Probability of making a sale (π)= 30% or 0.3

Part (a): Probability of making exactly four sales $(P(x = 4))$

$$\begin{aligned} P(x = 4) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{12} C_4 (0.30)^4 (1-0.30)^{12-4} \\ &= 0.231 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (b): Probability of making no sales $(P(x = 0))$

$$\begin{aligned} P(x = 0) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{12} C_0 (0.30)^4 (1-0.30)^{12-0} \\ &= 0.014 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (c): Probability of making exactly two sales $(P(x = 2))$

$$\begin{aligned} P(x = 2) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{12} C_2 (0.30)^2 (1-0.30)^{12-2} \\ &= 0.168 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (d): The mean number of sales in the two-hour period

$$\begin{aligned} \mu &= n\pi \\ &= 12 \times 0.30 \\ &= 3.6 \end{aligned}$$

Result

Part (a): 0.231

Part (b): 0.014

Part (c): 0.168

Part (d): 3.6

PROBLEM STATEMENT

Chapter 6, Page 202

Step 1

Number of graduates (n)=15

Probability of selecting public accounting (π)= 23% or 0.23

Part (a): Probability that two select public accounting $(P(x=2))$

$$\begin{aligned} P(x=2) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{15} C_2 (0.23)^2 (1-0.23)^{15-2} \\ &= 105 \times 0.053 \times (0.77)^{13} \\ &= 5.565 \times 0.0335 \\ &= 0.1864 \end{aligned}$$

Part (b): Probability thatfiveselect public accounting $(P(x=5))$

$$\begin{aligned} P(x=5) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{15} C_5 (0.23)^5 (1-0.23)^{15-5} \\ &= 3003 \times 0.00064 \times (0.77)^{10} \\ &= 1.933 \times 0.0734 \\ &= 0.142 \end{aligned}$$

Part (c): Expected number of graduates selecting public accounting (mean)

$$\begin{aligned} \mu &= n\pi \\ &= 15 \times 0.23 \\ &= 3.45 \end{aligned}$$

Result

Part (a): 0.1864

Part (b): 0.142

Part (c): 3.45

PROBLEM STATEMENT

Chapter 6, Page 202

Step 1

Number of households (n)= 8

Probability that a household uses a cell phone exclusively for telephone service (π)= 16% or 0.16

Part (a): Probability that none use a cell phone as their exclusive service $(P(x=0))$

$$\begin{aligned} P(x=0) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_8 C_0 (0.16)^0 (1-0.16)^{8-0} \\ &= 1 \times 1 \times (0.84)^8 \\ &= 0.248 \end{aligned}$$

Part (b): Probability that at least one uses the cell exclusively $(P(x=1))$

$$\begin{aligned} P(x=1) &= 1 - \text{ Probability that none use a cell phone as their exclusive service} \\ &= 1 - P(x=0) \\ &= 1 - 0.248 \\ &= 0.752 \end{aligned}$$

Part (c): Probability that at least five uses the cell exclusively $(P(x \geq 5))$

$$\begin{aligned} P(x \geq 5) &= P(5) + P(6) + P(7) + P(8) \\ &= {}_8 C_5 (0.16)^5 (1-0.16)^{8-5} + {}_8 C_6 (0.16)^6 (1-0.16)^{8-6} + {}_8 C_7 (0.16)^7 (1-0.16)^{8-7} + {}_8 C_8 (0.16)^8 (1-0.16)^{8-8} \\ &= (0.16)^5 ({}_8 C_5 (0.84)^3 + {}_8 C_6 (0.16)(0.84)^2 + {}_8 C_7 (0.16)^2 (0.84) + {}_8 C_8 (0.16)^3) \\ &= 0.0001(56 \times 0.593 + 28 \times 0.113 + 8 \times 0.0215 + 1 \times 0.0041) \\ &= 0.0001(33.208 + 3.164 + 0.175 + 0.0041) \\ &= 0.0001(36.5511) \\ &= 0.0036 \text{ which is almost } 0 \end{aligned}$$

Result

Part (a): 0.248

Part (b): 0.752

Part (c): 0

PROBLEM STATEMENT

Chapter 6, Page 203

Step 1

$n = 8$ and $\pi = 0.30$

Part (a)

$$\begin{aligned} P(x = 2) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_8 C_2 (0.30)^2 (1-0.30)^{8-2} \\ &= 0.296 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

Part (b)

$$\begin{aligned} P(x \le 2) &= P(x=0) + P(x=1) + P(x=2) \\ &= {}_8 C_0 (0.30)^0 (1-0.30)^{8-0} + {}_8 C_1 (0.30)^1 (1-0.30)^{8-1} + {}_8 C_2 (0.30)^2 (1-0.30)^{8-2} \\ &= 0.058 + 0.198 + 0.296 \text{ using the binomial probability distribution table given in appendix B} \\ &= 0.552 \end{aligned}$$

Part (c)

$$\begin{aligned} P(x \ge 3) &= 1 - P(x < 3) \\ &= 1 - P(x \le 2) \\ &= 1 - 0.552 \text{ from part } b \\ &= 0.448 \end{aligned}$$

Result

Part (a): 0.296

Part (b): 0.552

Part (c): 0.448

PROBLEM STATEMENT

Chapter 6, Page 204

Step 1

$n = 12$ and $\pi = 0.60$

a.

$$\begin{aligned} P(x=5) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= 12\ C5 (0.60)^5 (1-0.60)^{12-5} \\ &= 0.101 \text{ using the binomial probability distribution table given in appendix B} \end{aligned}$$

b.

$$\begin{aligned} P(x \le 5) =& P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5) \\ =& 12 C0 (0.60)^0 (1-0.60)^{12-0} + 12\ C1 (0.60)^1 (1-0.60)^{12-1} + 12\ C2 (0.60)^2 (1-0.60)^{12-2} \\ &+ 12\ C3 (0.60)^3 (1-0.60)^{12-3} + 12\ C4 (0.60)^4 (1-0.60)^{12-4} + 12\ C5 (0.60)^5 (1-0.60)^{12-5} \\ =& 0.000 + 0.000 + 0.002 + 0.012 + 0.042 + 0.101 \\ & \text{using the binomial probability distribution table given in appendix B} \\ =& 0.157 \end{aligned}$$

Part (c)

$$\begin{aligned} P(x \ge 6) &= 1 - P(x < 6) \\ &= 1 - P(x \le 5) \\ &= 1 - 0.157 \text{ from part } b \\ &= 0.843 \end{aligned}$$

Result

Part (a): 0.101

Part (b): 0.157

Part (c): 0.843

PROBLEM STATEMENT

Chapter 6, Page 204

Step 1

Sample of homes (n)=9

Probability that homes have large screen TVs (π)= 90% or 0.90

a. Probability that all nine have large screen TVs i.e $(P(x=9))$

$$\begin{aligned}P(x=9) &= {}_nC_x\pi^x(1-\pi)^{n-x}\\ &= {}_9C_9(0.90)^9(1-0.90)^{9-9}\\ &= 0.387 \text{ using the binomial probability distribution table given in appendix B}\end{aligned}$$

b. Probability that less than five have large-screen TVs i.e $(P(x\le 5))$

$$\begin{aligned}P(x<5) =& P(x=0)+P(x=1)+P(x=2)+P(x=3)+P(x=4)\\ =& {}_9C_0(0.90)^0(1-0.90)^{9-0} + {}_9C_1(0.90)^1(1-0.90)^{9-1} + {}_9C_2(0.90)^2(1-0.90)^{9-2}\\ &+ {}_9C_3(0.90)^3(1-0.90)^{9-3} + {}_9C_4(0.90)^4(1-0.90)^{9-4}\\ =& 0.000+0.000+0.000+0.000+0.001\\ &\text{using the binomial probability distribution table given in appendix B}\\ =& 0.001\end{aligned}$$

c. Probability that more than five have large-screen TVs i.e $(P(x>5))$

$$\begin{aligned}P(x>5) &= 1-P(x\le 5)\\ &= 1-(P(x<5)+P(x=5))\\ &= 1-(0.001+{}_9C_5(0.90)^5(1-0.90)^{9-5})\\ &= 0.999-0.007\\ &\text{using the binomial probability distribution table given in appendix B}\\ &= 0.992\end{aligned}$$

Step 2

d. Probability that at least seven homes have large-screen TVs i.e $(P(x\ge 7))$

$$\begin{aligned}P(x\ge 7) &= P(x=7)+P(x=8)+P(x=9)\\ &= {}_9C_7(0.90)^7(1-0.90)^{9-7} + {}_9C_8(0.90)^8(1-0.90)^{9-8} + {}_9C_9(0.90)^9(1-0.90)^{9-9})\\ &= 0.172+0.387+0.387\\ &\text{using the binomial probability distribution table given in appendix B}\\ &= 0.946\end{aligned}$$

Result

Part (a): 0.387

Part (b): 0.001

Part (c): 0.992

Part (d): 0.946

PROBLEM STATEMENT

Chapter 6, Page 204

Step 1

Number of window frames (n)= 20

Probability that the production will require an adjustment (π)= 5% or 0.05

Part (a): Probability that none will need adjustment $(P(x=0))$

$$\begin{aligned} P(x=0) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{20} C_0 (0.05)^0 (1-0.05)^{20-0} \\ &= 1 \times 1 \times (0.95)^{20} \\ &= 0.358 \end{aligned}$$

Part (b): Probability that at least one will need adjustment $(P(x \geq 1))$

$$\begin{aligned} P(x \geq 1) &= 1 - P(x < 1) \\ &= 1 - P(x=0) \\ &= 1 - 0.358 \\ &= 0.642 \end{aligned}$$

Part (c): Probability that more than two will need adjustment $(P(x > 2))$

$$\begin{aligned} P(x > 2) &= 1 - P(x \leq 2) \\ &= 1 - (P(x=0) + P(x=1) + P(x=2)) \\ &= 1 - ({}_{20}C_0 (0.05)^0 (1-0.05)^{20-0} + {}_{20} C_1 (0.05)^1 (1-0.05)^{20-1} + {}_{20} C_2 (0.05)^2 (1-0.05)^{20-2}) \\ &= 1 - (1 \times 1 \times (0.95)^{20} + 20 \times 0.05 \times (0.95)^{19} + 190 \times 0.025 \times (0.95)^{18}) \\ &= 1 - (0.358 + 0.377 + 0.189) \\ &= 0.076 \end{aligned}$$

Result

Part (a): 0.358

Part (b): 0.642

Part (c): 0.076

PROBLEM STATEMENT

Chapter 6, Page 204

Step 1

Number of cases reported (n)= 15

Probability that the problems can be resolved the same day (π): 70% or 0.7

a. Expected number of problems to be resolved the same day μ

$$\begin{aligned}\mu &= n\pi \\ &= 15 \times 0.7 \\ &= 10.5\end{aligned}$$

Standard deviation is

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{15 \times 0.7 \times (1-0.7)} \\ &= \sqrt{3.15} \\ &= 1.775\end{aligned}$$

$\textbf{b. Probability that 10 of the problems can be resolved today }$ i.e $(P(x=10))$

$$\begin{aligned}P(x=10) =& {}_n C_x \pi^x (1-\pi)^{n-x} \\ =& 15\ C10 (0.70)^{10} (1-0.70)^{15-10} \\ =& 0.206 \text{ using the binomial probability distribution table given in appendix B}\end{aligned}$$

$\textbf{c. Probability that 10 or 11 of the problems can be resolved today }$ i.e $(P(x=10))$ or $(P(x=10))$

$$\begin{aligned}P(x=10)+P(x=11) =& 15\ C10 (0.70)^{10} (1-0.70)^{15-10} + 15\ C11 (0.70)^{11} (1-0.70)^{15-11} \\ =& 0.206 + 0.219 \\ & \text{using the binomial probability distribution table given in appendix B} \\ =& 0.425\end{aligned}$$

Step 2

$\textbf{d. Probability that more than 10 of the problems can be resolved today }$ i.e $(P(x>10))$

$$\begin{aligned}P(x>10) =& P(x=11)+P(x=12)+P(x=13)+P(x=14)+P(x=15) \\ =& 15 C11 (0.70)^{11} (1-0.70)^{15-11} + 15\ C12 (0.70)^{12} (1-0.70)^{15-12} \\ & + 15\ C13 (0.70)^{13} (1-0.70)^{15-13} + 15\ C14 (0.70)^{14} (1-0.70)^{15-14} \\ & + 15\ C15 (0.70)^{15} (1-0.70)^{15-15} \\ =& 0.219 + 0.170 + 0.092 + 0.031 + 0.005 \\ & \text{using the binomial probability distribution table given in appendix B} \\ =& 0.517\end{aligned}$$

Result

Part (a): $\mu = 10.5$

Part (b): 0.206

Part (c): 0.425

Part (d): 0.517

PROBLEM STATEMENT

Chapter 6, Page 204

Step 1

Sample of cars (n)= 6

Probability that the car will have the transponder (π)=80% or 0.8

a. Probability that all six will have the transponder i.e $(P(x=6))$

$$\begin{aligned}P(x=6) =& {}_nC_x\pi^x(1-\pi)^{n-x}\\ =& {}_6C_6(0.80)^6(1-0.80)^{6-6}\\ =& \ 0.262 \text{ using the binomial probability distribution table given in appendix B}\end{aligned}$$

b. Probability that at least three will have the transponder i.e $(P(x\geq 3))$

$$\begin{aligned}P(x\geq 3) =& P(x=3)+P(x=4)+P(x=5)+P(x=6)\\ =& {}_6C_3(0.80)^3(1-0.80)^{6-3}+{}_6C_4(0.80)^4(1-0.80)^{6-4}\\ &+{}_6C_5(0.80)^5(1-0.80)^{6-5}+{}_6C_6(0.80)^6(1-0.80)^{6-6}\\ =& 0.082+0.246+0.393+0.262\\ & \text{using the binomial probability distribution table given in appendix B}\\ =& 0.983\end{aligned}$$

c. Probability that none will have the transponder i.e $(P(x=0))$

$$\begin{aligned}P(x=0) =& {}_nC_x\pi^x(1-\pi)^{n-x}\\ =& {}_6C_0(0.80)^0(1-0.80)^{6-0}\\ =& \ 0.000 \text{ using the binomial probability distribution table given in appendix B}\end{aligned}$$

Result

Part (a): 0.262

Part (b): 0.983

Part (c): 0.000

PROBLEM STATEMENT

Chapter 6, Page 207

Step 1

Total songs $N = 10$

Number of classical songs or successes (S)=6

Sample of songs (n)=3

Number of successes in the sample (x)= 2

Probability that exactly 2 songs are classical $(P(x = 2))$

$$\begin{aligned} P(x=2) &= \frac{(SC_x)(N-SC_{n-x})}{NC_n} \\ &= \frac{(6C_2)(6-2C_{3-2})}{10C_3} \\ &= \frac{15 \times 4}{120} \\ &= 0.5 \end{aligned}$$

Result

$$0.5$$

PROBLEM STATEMENT

Chapter 6, Page 207

Step 1

Population size $N = 15$

Number of acceptable items or successes (S)=10

Sample size (n)=4

Number of successes in the sample (x)= 3

Probability that exactly 3 are acceptable $(P(x = 3))$

$$\begin{aligned} P(x=3) &= \frac{(SC_x)(N-SC_{n-x})}{NC_n} \\ &= \frac{(10C_3)(15-10C_{4-3})}{15C_4} \\ &= \frac{120 \times 5}{1365} \\ &= 0.44 \end{aligned}$$

Result

$$0.44$$

PROBLEM STATEMENT

Chapter 6, Page 207

Step 1

Total DVD players $N = 10$

Number of defective units or successes (S)=3

Sample of DVD players (n)=2

Number of successes in the sample (x)= 0

Probability that neither of the 2 DVD players tested is defective $(P(x = 0))$

$$\begin{aligned} P(x = 0) &= \frac{(SC_x)(N-SC_{n-x})}{NC_n} \\ &= \frac{(3C_0)(10-3C_{2-0})}{10C_2} \\ &= \frac{1 \times 21}{45} \\ &= 0.467 \end{aligned}$$

Result

$$0.467$$

PROBLEM STATEMENT

Chapter 6, Page 207

Step 1

Total faculty members $N = 8$

Tenured faculty or successes (S)=6

Committee of faculty members (n)=3

Part (a): Number of successes in the sample (x)= 3

Probability that all members of the committee are tenured $(P(x = 3))$

$$\begin{aligned} P(x=3) &= \frac{(SC_x)(N-SC_{n-x})}{NC_n} \\ &= \frac{(6C_3)(8-6C_{3-3})}{8C_3} \\ &= \frac{20 \times 1}{56} \\ &= 0.36 \end{aligned}$$

Part (b): Let E be the event that at least one member is not tenured

$$\begin{aligned} P(E) &= 1 - P(\text{ none of the member is not tenured }) \\ &= 1 - P(\text{ all members are tenured }) \\ &= 1 - 0.36 \text{ from part } (a) \\ &= 0.64 \end{aligned}$$

Result

Part (a): 0.36

Part (b)=0.64

PROBLEM STATEMENT

Chapter 6, Page 207

Step 1

Total delivery trucks $N = 15$

Trucks with brake problems or successes (S)=6

Sample of trucks (n)=5

Number of successes in the sample (x)= 2

Probability that exactly 2 of trucks have defective brakes $(P(x = 2))$

$$\begin{aligned} P(x = 2) &= \frac{({}_{S}C_{x})({}_{N-S}C_{n-x})}{{}_{N}C_{n}} \\ &= \frac{({}_{6}C_{2})({}_{15-6}C_{5-2})}{{}_{15}C_{5}} \\ &= \frac{15 \times 84}{3003} \\ &= 0.419 \end{aligned}$$

Result

$$0.419$$

PROBLEM STATEMENT

Chapter 6, Page 207

Step 1

(a) The probability of matching all numbers is 1 in 3,838,380.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{all numbers match}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{1}{3,383,380} \\ &\approx 0.0000003 \\ &= 0.00003\% \end{aligned}$$

Thus there is a 0.00003% chance of matching all six numbers, which implies that it is very rare to mach all 6 numbers.

Step 2

(b) There are 6 winning numbers, while we draw 6 of the 40 numbers.

$$\begin{aligned} N &= \text{Population size} = 40 \\ n &= \text{Number of draws} = 6 \\ S &= \text{Number of observed successes} = 6 \end{aligned}$$

Formula **hypergeometric probability**:

$$P(X = k) = \frac{{}_{S}C_{X}\,{}_{N-S}C_{n-X}}{{}_{N}C_{n}}$$

Since we need to select 6 of the 6 winning numbers, we evaluate the definition of hypergeometric probability at $x = 6$:

$$\begin{aligned} P(X = 6) &= \frac{{}_{6}C_{6}\,{}_{40-6}C_{6-6}}{{}_{40}C_{6}} \\ &= \frac{{}_{6}C_{6}\,{}_{34}C_{0}}{{}_{40}C_{6}} \\ &= \frac{\frac{6!}{6!(6-6)!} \cdot \frac{34!}{0!(34-0)!}}{\frac{40!}{6!(40-6)!}} \\ &= \frac{1 \cdot 1}{3,838,380} \\ &= \frac{1}{3,838,380} \\ &\approx 0.0000003 \\ &= 0.00003\% \end{aligned}$$

Step 3

(c) There are 6 winning numbers, while we draw 6 of the 40 numbers.

$$\begin{aligned} N &= \text{Population size} = 40 \\ n &= \text{Number of draws} = 6 \\ S &= \text{Number of observed successes} = 6 \end{aligned}$$

Formula **hypergeometric probability**:

$$P(X = k) = \frac{{}_SC_X \, {}_{N-S}C_{n-X}}{{}_NC_n}$$

Since we need to select 4 of the 6 winning numbers, we evaluate the definition of hypergeometric probability at $x = 4$:

$$\begin{aligned} P(X = 4) &= \frac{{}_6C_4 \, {}_{40-6}C_{6-4}}{{}_{40}C_6} \\ &= \frac{{}_6C_4 \, {}_{34}C_2}{{}_{40}C_6} \\ &= \frac{\frac{6!}{4!(6-4)!} \cdot \frac{34!}{2!(34-2)!}}{\frac{40!}{6!(40-6)!}} \\ &= \frac{15 \cdot 561}{3,838,380} \\ &= \frac{8415}{3,838,380} \\ &\approx 0.0022 \\ &= 0.022\% \end{aligned}$$

Step 4

(d) There are 6 winning numbers, while we draw 6 of the 40 numbers.

$$\begin{aligned} N &= \text{Population size} = 40 \\ n &= \text{Number of draws} = 6 \\ S &= \text{Number of observed successes} = 6 \end{aligned}$$

Formula **hypergeometric probability**:

$$P(X = k) = \frac{{}_SC_X \, {}_{N-S}C_{n-X}}{{}_NC_n}$$

Since we need to select 5 of the 6 winning numbers, we evaluate the definition of hypergeometric probability at $x = 5$:

$$\begin{aligned} P(X = 5) &= \frac{{}_6C_5 \, {}_{40-6}C_{6-5}}{{}_{40}C_6} \\ &= \frac{{}_6C_5 \, {}_{34}C_1}{{}_{40}C_6} \\ &= \frac{\frac{6!}{5!(6-5)!} \cdot \frac{34!}{1!(34-1)!}}{\frac{40!}{6!(40-6)!}} \\ &= \frac{6 \cdot 34}{3,838,380} \\ &= \frac{204}{3,838,380} \\ &\approx 0.000053 \\ &= 0.0053\% \end{aligned}$$

Result

(a) $\frac{1}{3,383,380} \approx 0.0000003 = 0.00003\%$

(b) $\frac{1}{3,838,380} \approx 0.0000003 = 0.00003\%$

(c) $\frac{8415}{3,838,380} \approx 0.0022 = 0.022\%$

(d) $\frac{204}{3,838,380} \approx 0.000053 = 0.0053\%$

PROBLEM STATEMENT

Chapter 6, Page 212

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Step 2

SOLUTION

$$\mu=\text{Mean}=0.4$$

(a) Evaluate the probability of Poisson probability at $x=0$:

$$\begin{aligned}P(X=0)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{0.4^0 e^{-0.4}}{0!}\\&=\frac{1e^{-0.4}}{1}\\&=e^{-0.4}\\&\approx 0.6703\\&=67.03\%\end{aligned}$$

Step 3

(b) Use the complement rule and the result of part (a):

$$\begin{aligned}P(X>0)&=1-P(X=0)\\&=1-e^{-0.4}\\&\approx 0.3297\\&=32.97\%\end{aligned}$$

Result

(a) $e^{-0.4}\approx 0.6703=67.03\%$

(b) $1-e^{-0.4}\approx 0.3297=32.97\%$

PROBLEM STATEMENT

Chapter 6, Page 212

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Step 2

SOLUTION

$$\mu=\text{Mean}=4$$

(a) Evaluate the probability of Poisson probability at $x=2$:

$$\begin{aligned}P(X=2)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{4^2e^{-4}}{2!}\\&=\frac{16e^{-4}}{2}\\&=8e^{-4}\\&\approx 0.1465\\&=14.65\%\end{aligned}$$

Step 3

(b) Evaluate the probability of Poisson probability at $x = 0, 1$:

$$\begin{aligned} P(X=0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{4^0 e^{-4}}{0!} \\ &= \frac{1e^{-4}}{1} \\ &= e^{-4} \\ &\approx 0.0183 \\ P(X=1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{4^1 e^{-4}}{0!} \\ &= \frac{4e^{-4}}{1} \\ &= 4e^{-4} \\ &\approx 0.0733 \end{aligned}$$

Use the addition rule for mutually exclusive events along with the result of part (a):

$$\begin{aligned} P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= 0.0183 + 0.0733 + 0.1465 \\ &= 0.2381 \\ &= 23.81\% \end{aligned}$$

Step 4

(c) Use the complement rule and the result of part (b):

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - 0.2381 \\ &= 0.7619 \\ &= 76.19\% \end{aligned}$$

Result

(a) $8e^{-4} \approx 0.1465 = 14.65\%$

(b) $0.2381 = 23.81\%$

(c) $0.7619 = 76.19\%$

PROBLEM STATEMENT

Chapter 6, Page 212

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\begin{aligned} n &= \text{Sample size} = 40 \\ \pi &= \text{Probability of success} = 0.025 \end{aligned}$$

(a) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 40(0.025) = 1$$

Evaluate the probability of Poisson probability at $x=3$:

$$\begin{aligned} P(X=3) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{1^3 e^{-1}}{3!} \\ &= \frac{1e^{-1}}{6} \\ &= \frac{1}{6}e^{-1} \\ &\approx 0.0613 \\ &= 6.13\% \end{aligned}$$

Step 3

(b) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 40(0.025) = 1$$

Evaluate the probability of Poisson probability at $x = 0, 1, 2$:

$$\begin{aligned} P(X=0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{1^0 e^{-1}}{0!} \\ &= \frac{1e^{-1}}{1} \\ &= e^{-1} \\ &\approx 0.3679 \\ P(X=1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{1^1 e^{-1}}{1!} \\ &= \frac{1e^{-1}}{1} \\ &= e^{-1} \\ &\approx 0.3679 \\ P(X=2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{1^2 e^{-1}}{2!} \\ &= \frac{1e^{-1}}{2} \\ &= \frac{1}{2}e^{-1} \\ &\approx 0.1839 \end{aligned}$$

Step 4

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same month):

$$\begin{aligned} P(X<3) &= P(X=0) + P(X=1) + P(X=2) \\ &= 0.3679 + 0.3679 + 0.1839 \\ &= 0.9197 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 3) &= 1 - P(X<3) \\ &= 1 - 0.9197 \\ &= 0.0803 \\ &= 8.03\% \end{aligned}$$

Result

(a) $\frac{1}{0}e^{-1} \approx 0.0613 = 6.13\%$

(b) $0.0803 = 8.03\%$

PROBLEM STATEMENT

Chapter 6, Page 212

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Step 2

SOLUTION

$$\mu = \text{Mean} = 2 \text{ (per minute)}$$

(a) Evaluate the probability of Poisson probability at $x = 0$:

$$\begin{aligned} P(X = 0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2^0 e^{-2}}{0!} \\ &= \frac{1e^{-2}}{1} \\ &= e^{-2} \\ &\approx 0.1353 \\ &= 13.53\% \end{aligned}$$

Step 3

(b) Use the complement rule and the result of part (a):

$$\begin{aligned} P(X \geq 1) &= P(X > 0) \\ &= 1 - P(X = 0) \\ &= 1 - e^{-2} \\ &\approx 0.8647 \\ &= 86.47\% \end{aligned}$$

Result

(a) $e^{-2} \approx 0.1353 = 13.53\%$

(b) $1 - e^{-2} \approx 0.8647 = 86.47\%$

PROBLEM STATEMENT

Chapter 6, Page 212

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\begin{aligned} n &= \text{Sample size} = 1200 \\ \pi &= \text{Probability of success} = 0.5\% = 0.005 \end{aligned}$$

The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 1200(0.005) = 6$$

Evaluate the probability of Poisson probability at $x=0,1,2,3,4$:

$$\begin{aligned} P(X=0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{6^0 e^{-6}}{0!} \\ &= \frac{6e^{-6}}{1} \\ &= e^{-6} \\ &\approx 0.0025 \\ P(X=1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{6^1 e^{-6}}{1!} \\ &= \frac{6e^{-6}}{1} \\ &= 6e^{-6} \\ &\approx 0.0149 \end{aligned}$$

Step 3

$$
\begin{aligned}
P(X=2) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{6^2 e^{-6}}{2!} \\
&= \frac{36e^{-6}}{2} \\
&= 18e^{-6} \\
&\approx 0.0446 \\
P(X=3) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{6^3 e^{-6}}{3!} \\
&= \frac{216e^{-6}}{6} \\
&= 36e^{-6} \\
&\approx 0.0892 \\
P(X=4) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{6^4 e^{-6}}{4!} \\
&= \frac{1296e^{-6}}{24} \\
&= 54e^{-6} \\
&\approx 0.1338
\end{aligned}
$$

Step 4

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same day):

$$
\begin{aligned}
P(X<5) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) \\
&= 0.0025 + 0.0149 + 0.0446 + 0.0892 + 0.1338 \\
&= 0.2850
\end{aligned}
$$

Use the complement rule:

$$
\begin{aligned}
P(X \geq 5) &= 1 - P(X<5) \\
&= 1 - 0.2850 \\
&= 0.7150 \\
&= 71.50\%
\end{aligned}
$$

🎯 Result

$$0.7150 = 71.50\%$$

PROBLEM STATEMENT

Chapter 6, Page 212

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\mu=\text{Mean}=3 \text{ (per year)}$$

Evaluate the probability of Poisson probability at $x=4$:

$$\begin{aligned}P(X=4)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{3^4e^{-3}}{4!}\\&=\frac{81e^{-3}}{24}\\&=\frac{27}{8}e^{-3}\\&\approx 0.1680\\&=16.80\%\end{aligned}$$

Result

$$\frac{27}{8}e^{-3}\approx 0.1680=16.80\%$$

PROBLEM STATEMENT

Chapter 6, Page 213

Step 1

A **random variable** is a variable that is assigned a value at random from some set of possible values.

A **probability distribution** is a function that assigns a probability (value between 0 and 1) to all possible values of a random variable.

Thus we note that a probability distribution includes a probability besides the possible values of a random variable, while a random variable contains only the possible values.

Result

A probability distribution includes a probability besides the possible values of a random variable, while a random variable contains only the possible values.

PROBLEM STATEMENT

Chapter 6, Page 213

Step 1

Discrete data are restricted to defined separated values, for example integers or counts. **Continuous** data are not restricted to defined separate values, but can occupy any value over a continuous range, for example decimal, rational or real numbers.

(a) Continuous, because the length of time can take on decimal values (such as 27.5 minutes).

(b) Discrete, because the number of cars are counts (that is, nonnegative integers).

(c) Discrete, because the number of hits are counts (that is, nonnegative integers).

(d) Discrete, because the number of patients are counts (that is, nonnegative integer).

(e) Continuous, because the distance can take on decimal values (such as 7.5 miles or 9.75 kilometers).

(f) Discrete, because the number of customers are counts (that is, nonnegative integer).

(g) Continuous, because the distance can take on decimal values (such as 7.5 miles or 9.75 kilometers).

Result

(a) Continuous

(b) Discrete

(c) Discrete

(d) Discrete

(e) Continuous

(f) Discrete

(g) Continuous

PROBLEM STATEMENT

Chapter 6, Page 213

Step 1

The mean of the worth of the investment is

$$\begin{aligned}\mu &= \Sigma[xP(x)]\\ \mu &= \$1000 \times 0.25 + \$2000 \times 0.60 + \$5000 \times 0.15\\ \mu &= \$250 + \$1200 + \$750\\ \mu &= \$2200\end{aligned}$$

The formula for the variance is

$$\begin{aligned}\sigma^2 &= \Sigma[(x-\mu)^2 P(x)]\\ \sigma^2 &= (1000-2200)^2 \times 0.25 + (2000-2200)^2 \times 0.60 + (5000-2200)^2 \times 0.15\\ \sigma^2 &= 1440000 \times 0.25 + 40000 \times 0.60 + 7840000 \times 0.15\\ \sigma^2 &= 360000 + 24000 + 1176000\\ \sigma^2 &= 1560000\end{aligned}$$

Result

$\mu = \$2200$ and $\sigma^2 = 1560000$

PROBLEM STATEMENT

Chapter 6, Page 213

Step 1

The mean number of accidents is

$$\begin{aligned}\mu &= \Sigma[xP(x)]\\ \mu &= 0\times 0.4+1\times 0.2+2\times 0.2+3\times 0.1+4\times 0.1\\ \mu &= 0+0.2+0.4+0.3+0.4\\ \mu &= 1.3\end{aligned}$$

The variance of the number of accidents is

$$\begin{aligned}\sigma^2 &= \Sigma[(x-\mu)^2P(x)]\\ \sigma^2 &= (0-1.3)^2\times 0.4+(1-1.3)^2\times 0.2+(2-1.3)^2\times 0.2+(3-1.3)^2\times 0.1+(4-1.3)^2\times 0.1\\ \sigma^2 &= 1.69\times 0.4+0.09\times 0.2+0.49\times 0.2+2.89\times 0.1+7.29\times 0.1\\ \sigma^2 &= 0.676+0.018+0.098+0.289+0.729\\ \sigma^2 &= 1.81\end{aligned}$$

The standard deviation is

$$\begin{aligned}\sigma &= \sqrt{\sigma^2}\\ &= \sqrt{1.81}\\ &= 1.345\end{aligned}$$

Result

$\mu = 1.3$, $\sigma^2 = 1.81$ and $\sigma = 1.345$

PROBLEM STATEMENT

Chapter 6, Page 214

Step 1

Let x be the number of cakes sold and $P(x)$ be the probability of x cakes sold

The mean of the number of cakes sold is

$$\begin{aligned}\mu &= \Sigma[xP(x)]\\ \mu &= 12 \times 0.25 + 13 \times 0.4 + 14 \times 0.25 + 15 \times 0.1\\ \mu &= 3 + 5.2 + 3.5 + 1.5\\ \mu &= 13.2\end{aligned}$$

The variance of the number of cakes sold is

$$\begin{aligned}\sigma^2 &= \Sigma[(x-\mu)^2 P(x)]\\ \sigma^2 &= (12-13.2)^2 \times 0.25 + (13-13.2)^2 \times 0.4 + (14-13.2)^2 \times 0.25 + (15-13.2)^2 \times 0.1\\ \sigma^2 &= 1.44 \times 0.25 + 0.04 \times 0.4 + 0.64 \times 0.25 + 3.24 \times 0.1\\ \sigma^2 &= 0.36 + 0.016 + 0.16 + 0.324\\ \sigma^2 &= 0.86\end{aligned}$$

The standard deviation is

$$\begin{aligned}\sigma &= \sqrt{\sigma^2}\\ &= \sqrt{0.86}\\ &= 0.93\end{aligned}$$

Result

$\mu = 13.2$, $\sigma^3 = 0.86$ and $\sigma = 0.93$

PROBLEM STATEMENT

Chapter 6, Page 214

Step 1

Given:

Payout x	Probability $P(X = x)$
50,000,000	0.000000006844
200,000	0.000000280614
10,000	0.000001711060
100	0.000070145903
100	0.000083836351
7	0.003424657534
7	0.001340482574
4	0.007812500000
3	0.014285714286
-1	0.9729806648

Note that the payout of -1 is obtained when you do't win anything and its probability is 1 decreased by all other known probabilities.

The expected value (or mean) is the sum of the product of each possibility x with its probability $P(X = x)$:

$$\begin{aligned}\mu &= \sum xP(X = x)\\ &= 50,000,000 \times 0.000000006844 + 200,000 \times 0.000000280614\\ &+ 10,000 \times 0.000001711060 + 100 \times 0.000070145903\\ &+ 100 \times 0.000083836351 + 7 \times 0.003424657534\\ &+ 7 \times 0.001340482574 + 4 \times 0.007812500000\\ &+ 3 \times 0.014285714286 + (-1) \times 0.9729806648\\ &\approx -0.43\end{aligned}$$

Step 2

The variance is the expected value of the squared deviation from the mean:

$$\begin{aligned}\sigma^2 &= \sum (x-\mu)^2 P(x)\\ &= (50,000,000-(-0.43))^2 \times 0.000000006844\\ &+ (200,000-(-0.43))^2 \times 0.000000280614\\ &+ (10,000-(-0.43))^2 \times 0.000001711060\\ &+ (100-(-0.43))^2 \times 0.000070145903\\ &+ (100-(-0.43))^2 \times 0.000083836351 + (7-(-0.43))^2 \times 0.003424657534\\ &+ (7-(-0.43))^2 \times 0.001340482574 + (4-(-0.43))^2 \times 0.007812500000\\ &+ (3-(-0.43))^2 \times 0.014285714286 + ((-1)-(-0.43))^2 \times 0.9729806648\\ &\approx 17,121,398.48\end{aligned}$$

The standard deviation is the square root of the variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{17,121,398.48} \approx 4137.80$$

Result

Mean $-\$0.43$ and Standard deviation $\$4137.80$

PROBLEM STATEMENT

Chapter 6, Page 214

Step 1

Sample size of people (n)= 10

Probability that chocolate was favorite flavor of ice cream (π)= 35% or 0.35

a. Expected number of people having chocolate as favorite flavor μ

$$\begin{aligned}\mu &= n\pi \\ &= 10 \times 0.35 \\ &= 3.5\end{aligned}$$

b. Probability that exactly four of those in the sample name chocolate i.e $(P(x=4))$

$$\begin{aligned}P(x=4) =& {}_n\,C_x \pi^x (1-\pi)^{n-x} \\ =& {}_{10}\,C_4 (0.35)^4 (1-0.35)^{9-4} \\ =& 210 \times 0.015 \times (0.65)^5 \\ =& 3.15 \times 0.116 \\ =& 0.3654\end{aligned}$$

c. Probability that four or more name chocolate i.e $(P(x \geq 4))$

$$\begin{aligned}P(x \geq 4) =& 1 - P(x < 4) \\ =& 1 - (P(x=0) + P(x=1) + P(x=2) + P(x=3)) \\ =& 1 - ({}_{10}C_0 (0.35)^0 (1-0.35)^{9-0} + {}_{10}\,C_1 (0.35)^1 (1-0.35)^{9-1} \\ & + {}_{10}\,C_2 (0.35)^2 (1-0.35)^{9-2} + {}_{10}\,C_3 (0.35)^3 (1-0.35)^{9-3}) \\ =& 1 - (1(1)(0.65)^9 + 10(0.35)(0.65)^8 \\ & + 45(0.1225)(0.65)^7 + 120(0.043)((0.65)^6) \\ =& 1 - (0.0207 + 0.1115 + 0.2702 + 0.389) \\ =& 1 - 0.7914 \\ =& 0.2086\end{aligned}$$

Result

Part (a): 3.5

Part (b): 0.3654

Part (c): 0.2086

PROBLEM STATEMENT

Chapter 6, Page 214

Step 1

Given:

$$n = \text{Sample size} = 12$$
$$\pi = \text{Probability of success} = 30\% = 0.30$$

The number of successes among a fixed number of independent trials follows a **binomial distribution**.

Assumptions

The four conditions for a binomial setting are: binary (success/failure), independent trials, fixed number of trials and probability of success is the same for each trial.

Binary: Satisfied, because Success=Spanish-speaking and Failure=

Not Spanish-speaking.

Independent trials: Satisfied, assuming that the potential jurors are randomly selected.

Fixed number of trials: Satisfied, because we selected 12 potential jurors.

Probability of success: Satisfied, because there is a 30% chance of an American being Spanish-speaking and thus the probability of success is 30%.

Since all 4 conditions are satisfied, the given scenario describes a binomial setting.

Step 2

Probability

Definition **binomial probability**:

$$P(X = k) = {}_n C_k \cdot \pi^k \cdot (1-\pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1-\pi)^{n-k}$$

Evaluate the definition of binomial probability at $k = 2$:

$$\begin{aligned} P(X = 2) &= {}_{12} C_2 \cdot 0.30^2 \cdot (1-0.30)^{12-2} \\ &= \frac{12!}{2!(12-2)!} \cdot 0.30^2 \cdot 0.70^{10} \\ &= 66 \cdot 0.30^2 \cdot 0.70^{10} \\ &\approx 0.1677 \\ &= 16.77\% \end{aligned}$$

We note that the probability of obtaining 2 Spanish-speaking Americans is large (greater than 5%), which implies that it is not unusual to obtain only 2 Spanish-speaking Americans among 12 potential jurors and thus there does not appear to be bias against her client.

Result

No bias

Assumption: The potential jurors are randomly selected

PROBLEM STATEMENT

Chapter 6, Page 214

Step 1

Sample size of policyholder (n)= 15

Probability that policyholders submit a claim during the year (π)= 40% or 0.4

a. Expected number of policyholders to file a claim μ

$$\begin{aligned}\mu &= n\pi \\ &= 15 \times 0.4 \\ &= 6\end{aligned}$$

b. Probability that 10 of the selected policyholders submitted a claim i.e $(P(x = 10))$

$$\begin{aligned}P(x = 10) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}_{15} C_{10} (0.40)^{10} (1-0.40)^{15-10} \\ &= 0.024 \text{ using the binomial probability distribution table given in appendix B}\end{aligned}$$

c. Probability that 10 or more of the selected policyholders submitted a claim i.e $(P(x \geq 10))$

$$\begin{aligned}P(x \geq 10) &= P(x = 10) + P(x = 11) + P(x = 12) + P(x = 13) + P(x = 14) + P(x = 15) \\ &= 0.024 + 0.007 + 0.002 + 0.000 + 0.000 + 0.000 \\ &\text{(using the binomial probability distribution table given in appendix B)} \\ &= 0.033\end{aligned}$$

c. Probability that more than 10 of the selected policyholders submitted a claim i.e $(P(x > 10))$

$$\begin{aligned}P(x > 10) &= P(x = 11) + P(x = 12) + P(x = 13) + P(x = 14) + P(x = 15) \\ &= 0.007 + 0.002 + 0.000 + 0.000 + 0.000 \\ &\text{(using the binomial probability distribution table given in appendix B)} \\ &= 0.009\end{aligned}$$

Result

Part (a): 6

Part (b): 0.024

Part (c): 0.033

Part (d): 0.009

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

Given:

$$n = \text{Sample size} = 18$$
$$\pi = \text{Probability of success} = \frac{2}{3} \approx 0.6667$$

The number of successes among a fixed number of independent trials follows a **binomial distribution**.

Definition **binomial probability**:

$$P(X = k) = {}_n C_k \cdot \pi^k \cdot (1-\pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1-\pi)^{n-k}$$

Evaluate the definition of binomial probability at $k = 14$:

$$\begin{aligned} P(X = 14) &= {}_{18} C_{14} \cdot 0.6667^{14} \cdot (1 - 0.6667)^{18-14} \\ &= \frac{18!}{14!(18-14)!} \cdot 0.6667^{14} \cdot 0.3333^4 \\ &= 3060 \cdot 0.6667^{14} \cdot 0.3333^4 \\ &\approx 0.1294 \\ &= 12.94\% \end{aligned}$$

Result

$$0.1294 = 12.94\%$$

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

Sample size of employed workers (n)= 20

Probability that an employee has a drug problem (π)= 7.5% or 0.075

Part (a): Expected number of employees having a drug problem

$$\begin{aligned}\mu &= n\pi \\ &= 20 \times 0.075 \\ &= 1.5\end{aligned}$$

Standard deviation is

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{20 \times 0.075 \times (1-0.075)} \\ &= 1.178\end{aligned}$$

Part (b): Probability that none of the workers sampled has a drug problem $(P(x=0))$

$$\begin{aligned}P(x=0) &= {}_nC_x \pi^x (1-\pi)^{n-x} \\ &= {}_{20}C_0 (0.075)^0 (1-0.075)^{20-0} \\ &= 1 \times 1 \times (0.925)^{20} \\ &= 0.2103\end{aligned}$$

Part (c): Probability that at least one has a drug problem $(P(x \geq 1))$

$$\begin{aligned}P(x \geq 1) &= 1 - P(x < 1) \\ &= 1 - P(x=0) \\ &= 1 - 0.2103 \text{ from part } (a) \\ &= 0.7897\end{aligned}$$

Result

Part (a): $\mu = 1.5$ and $\sigma = 1.178$

Part (b): 0.2103

Part (c): 0.7897

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

New cards (n)= 12

Probability that a credit card holder will default (π)= 7% or 0.07

Part (a): Expected number of card holders to default

$$\begin{aligned}\mu &= n\pi \\ &= 12 \times 0.07 \\ &= 0.84\end{aligned}$$

Standard deviation is

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{12 \times 0.07 \times (1-0.07)} \\ &= 0.884\end{aligned}$$

Part (b): Probability that none of the cardholders will default $(P(x=0))$

$$\begin{aligned}P(x=0) &= {}_n C_x \pi^x (1-\pi)^{n-x} \\ &= {}^{12} C_0 (0.07)^0 (1-0.07)^{12-0} \\ &= 1 \times 1 \times (0.93)^{12} \\ &= 0.418\end{aligned}$$

Part (c): Probability that at least one will default $(P(x \geq 1))$

$$\begin{aligned}P(x \geq 1) &= 1 - P(x < 1) \\ &= 1 - P(x=0) \\ &= 1 - 0.418 \text{ from part } (a) \\ &= 0.582\end{aligned}$$

Result

Part (a): $\mu = 0.84$ and $\sigma = 0.884$

Part (b): 0.418

Part (c): 0.582

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

Sample size (n)= 16

Probability of a person who visits a retail site on the World Wide Web make a purchase (π): 15% or 0.15

a. Probability of exactly four purchases i.e $P(x=4)$

$$\begin{aligned}P(x=4) =& {}_n C_x \pi^x (1-\pi)^{n-x} \\ =& {}_{16} C_4 (0.15)^4 (1-0.15)^{16-4} \\ =& 1820 \times 0.0005 \times (0.85)^{12} \\ =& 0.91 \times 0.1422 \\ =& 0.1294\end{aligned}$$

Step 2

b. Expected number of purchases μ

$$\begin{aligned}\mu &= n\pi \\ &= 16 \times 0.15 \\ &= 2.4\end{aligned}$$

Step 3

c. Probability that four or more hits result in a purchase $P(x \geq 4)$

$$\begin{aligned}P(x \geq 4) &= 1 - P(x < 4) \\ =& 1 - (P(x=0) + P(x=1) + P(x=2) + P(x=3)) \\ =& 1 - ({}_{16}C_0 (0.15)^0 (1-0.15)^{16-0} + {}_{16} C_1 (0.15)^1 (1-0.15)^{16-1} \\ & + {}_{16} C_2 (0.15)^2 (1-0.15)^{16-2} + {}_{16} C_3 (0.15)^3 (1-0.15)^{16-3}) \\ =& 1 - (1 \times 1 \times (0.85)^{16} + 16 \times 0.15 \times (0.85)^{15} \\ & + 120 \times 0.0225 \times (0.85)^{14} + 560 \times 0.0034 \times (0.85)^{13}) \\ =& 1 - (0.074 + 0.2096 + 0.2774 + 0.2302) \\ =& 1 - (0.7912) \\ =& 0.2088\end{aligned}$$

Result

Part (a): 0.1294

Part (b): 2.4

Part (c): 0.2088

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

DEFINITIONS

Definition **binomial probability**:

$$P(X=k) = {}_n C_k \cdot \pi^k \cdot (1-\pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1-\pi)^{n-k}$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Step 2

SOLUTION

(a) Given:

$$\begin{aligned} n &= \text{Sample size} = 20 \\ \pi &= \text{Probability of success} = 1\% = 0.01 \end{aligned}$$

The number of successes among a fixed number of independent trials follows a **binomial distribution**.

The lot is accepted when it contains 0 defects or 1 defect.

Evaluate the definition of binomial probability at $k=0$ and $k=1$:

$$\begin{aligned} P(X=0) &= {}_{20} C_0 \cdot 0.01^0 \cdot (1-0.01)^{20-0} \\ &= \frac{20!}{0!(20-0)!} \cdot 0.01^0 \cdot 0.99^{20} \\ &= 1 \cdot 1 \cdot 0.99^{20} \\ &\approx 0.8179 \\ P(X=1) &= {}_{20} C_1 \cdot 0.01^1 \cdot (1-0.01)^{20-1} \\ &= \frac{20!}{1!(20-0)!} \cdot 0.01^1 \cdot 0.99^{19} \\ &= 20 \cdot 0.01 \cdot 0.99^{19} \\ &\approx 0.1652 \end{aligned}$$

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of success among the same set of trials).

$$\begin{aligned} P(\text{accept}) &= P(X \le 1) \\ &= P(X=0) + P(X=1) \\ &= 0.8179 + 0.1652 \\ &= 0.9831 \\ &= 98.31\% \end{aligned}$$

Step 3

(b) Given:

$$\begin{aligned} n &= \text{Sample size} = 20 \\ \pi &= \text{Probability of success} = 2\% = 0.02 \end{aligned}$$

The number of successes among a fixed number of independent trials follows a **binomial distribution**.

The lot is accepted when it contains 0 defects or 1 defect.

Evaluate the definition of binomial probability at $k = 0$ and $k = 1$:

$$\begin{aligned} P(X = 0) &= {}_{20}C_0 \cdot 0.02^0 \cdot (1 - 0.02)^{20-0} \\ &= \frac{20!}{0!(20-0)!} \cdot 0.02^0 \cdot 0.98^{20} \\ &= 1 \cdot 1 \cdot 0.98^{20} \\ &\approx 0.6676 \\ P(X = 1) &= {}_{20}C_1 \cdot 0.02^1 \cdot (1 - 0.02)^{20-1} \\ &= \frac{20!}{1!(20-0)!} \cdot 0.02^1 \cdot 0.98^{19} \\ &= 20 \cdot 0.02 \cdot 0.98^{19} \\ &\approx 0.2725 \end{aligned}$$

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of success among the same set of trials).

$$\begin{aligned} P(\text{accept}) &= P(X \le 1) \\ &= P(X = 0) + P(X = 1) \\ &= 0.6676 + 0.2725 \\ &= 0.9401 \\ &= 94.01\% \end{aligned}$$

Step 4

(c) Given:

$$\begin{aligned} n &= \text{Sample size} = 20 \\ \pi &= \text{Probability of success} = 5\% = 0.05 \end{aligned}$$

The number of successes among a fixed number of independent trials follows a **binomial distribution**.

The lot is accepted when it contains 0 defects or 1 defect.

Evaluate the definition of binomial probability at $k = 0$ and $k = 1$:

$$\begin{aligned} P(X = 0) &= {}_{20}C_0 \cdot 0.05^0 \cdot (1 - 0.05)^{20-0} \\ &= \frac{20!}{0!(20-0)!} \cdot 0.05^0 \cdot 0.95^{20} \\ &= 1 \cdot 1 \cdot 0.95^{20} \\ &\approx 0.3585 \\ P(X = 1) &= {}_{20}C_1 \cdot 0.05^1 \cdot (1 - 0.05)^{20-1} \\ &= \frac{20!}{1!(20-0)!} \cdot 0.05^1 \cdot 0.95^{19} \\ &= 20 \cdot 0.05 \cdot 0.95^{19} \\ &\approx 0.2725 \end{aligned}$$

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of success among the same set of trials).

$$\begin{aligned} P(\text{accept}) &= P(X \le 1) \\ &= P(X = 0) + P(X = 1) \\ &= 0.3585 + 0.3774 \\ &= 0.7359 \\ &= 73.59\% \end{aligned}$$

Result

(a) $0.9831 = 98.31\%$

(b) $0.9401 = 94.01\%$

(c) $0.7359 = 73.59\%$

Chapter 6 **Exercise 51** Page 215

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

Population size $N = 10$

People who liked the new flavor or number of successes (S)=6

Sample size (n)=4

Number of successes in the sample (x)= 2

Probability that two liked the new flavor $(P(x = 2))$

$$\begin{aligned} P(x = 2) &= \frac{(SCx)(N-SCn-x)}{NCn} \\ &= \frac{(6C2)(10-6C4-2)}{10C4} \\ &= \frac{15 \times 6}{210} \\ &= 0.43 \end{aligned}$$

Result

0.13

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

Probability that a college student watches soap operas (π): 45% or 0.45

Sample size of college students(n): 10

a. Probability distribution of the students who watch soap operas

Let x be the number of students who watch soap operas where $x =, 1, 2, ..., 10$. Then probability that x students watch soap operas is $P(x)$

$$\begin{aligned} P(x) =& {}_n\, C_x \pi^x (1-\pi)^{n-x} \\ =& {}_{10}\, C_x (0.45)^x (1-0.45))^{10-x} \\ =& {}_{10}\, C_x (0.45)^x (0.55)^{10-x} \end{aligned}$$

b. Mean of this distribution is μ

$$\begin{aligned} \mu &= n\pi \\ &= 10 \times 0.45 \\ &= 4.5 \end{aligned}$$

Standard deviation is

$$\begin{aligned} \sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{10 \times 0.45 \times (1-0.45)} \\ &= \sqrt{2.475} \\ &= 1.573 \end{aligned}$$

c. Probability that exactly four students watch soap operas i.e $(P(x=4))$

$$\begin{aligned} P(x=4) =& {}_n\, C_x \pi^x (1-\pi)^{n-x} \\ =& {}_{10}\, C_4 (0.45)^4 (0.55)^{10-4} \\ =& \ 210 \times 0.041 \times 0.0277 \\ =& \ 0.238 \end{aligned}$$

Step 2

d. Probability that less than half of the students selected watch soap operas i.e $(P(x<5))$

$$\begin{aligned} P(x<5) &= P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) \\ &= {}_{10}\, C_0 (0.45)^0 (0.55)^{10-0} + {}_{10}\, C_1 (0.45)^1 (0.55)^{10-1} + {}_{10}\, C_2 (0.45)^2 (0.55)^{10-2} \\ &\quad + {}_{10}\, C_3 (0.45)^3 (0.55)^{10-3} + {}_{10}\, C_4 (0.45)^4 (0.55)^{10-4} \\ &= 1 \times 1 \times 0.0025 + 10 \times 0.45 \times 0.0046 + 45 \times 0.2025 \times 0.0084 \\ &\quad + 120 \times 0.091 \times 0.0152 + 210 \times 0.041 \times 0.0277 \\ &= 0.0025 + 0.0207 + 0.0765 + 0.166 + 0.238 \\ &= 0.5037 \end{aligned}$$

Result

Part (a): $P(x) = {}_{10}\, C_x (0.45)^x (0.55)^{10-x}$

Part (b): $\mu = 4.5$ and $\sigma = 1.573$

Part (c): 0.238

Part (d): 0.5037

PROBLEM STATEMENT

Chapter 6, Page 215

Step 1

Sample size of business travelers (n)=12

Probability that business travelers plan their trips less than two weeks before departure (π)= 52% or 0.52

a. The probability distribution for the number of travelers who plan their trips within two weeks of departure

Let (x) be the number of travelers who plan their trips within two weeks of departure where $x = 0, 1, 2, ..., 12.$

Probability that x travelers plan their trips before departure is $P(x)$

$$\begin{aligned} P(x) =& {}_n C_x \pi^x (1-\pi)^{n-x} \\ =& {}_{12} C_x (0.52)^x (1-0.52)^{12-x} \\ =& {}_{12} C_x (0.52)^x (0.48)^{12-x} \end{aligned}$$

b. Mean of this distribution μ

$$\begin{aligned} \mu &= n\pi \\ &= 12 \times 0.52 \\ &= 6.24 \end{aligned}$$

The standard deviation of the distribution is

$$\begin{aligned} \sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{12 \times 0.52 \times (1-0.52)} \\ &= \sqrt{2.9952} \\ &= 1.73 \end{aligned}$$

c. Probability that exactly 5 of the 12 selected business travelers plan their trips within two weeks of departure i.e $(P(x = 5))$

$$\begin{aligned} P(x=5) =& {}_{12} C_5 (0.52)^5 (0.48)^{12-5} \\ =& 792 \times 0.038 \times 0.0059 \\ =& 0.1775 \end{aligned}$$

Step 2

d. Probability 5 or fewer of the 12 selected business travelers plan their trips within two weeks of departure i., $P(x \leq 5)$

$$\begin{aligned} P(x \leq 5) =& P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5) \\ =& {}_{12} C_0 (0.52)^0 (0.48)^{12-0} + {}_{12} C_1 (0.52)^1 (0.48)^{12-1} + {}_{12} C_2 (0.52)^2 (0.48)^{12-2} \\ & + {}_{12} C_3 (0.52)^3 (0.48)^{12-3} + {}_{12} C_4 (0.52)^4 (0.48)^{12-4} + {}_{12} C_5 (0.52)^5 (0.48)^{12-5} \\ =& 1 \times 1 \times 0.00015 + 12 \times 0.52 \times 0.00031 + 66 \times 0.2704 \times 0.00065 + 220 \times 0.141 \times 0.00135 \\ & + 495 \times 0.0731 \times 0.028 + 792 \times 0.038 \times 0.0059 \\ =& 0.00015 + 0.0019 + 0.0116 + 0.042 + 0.1775 \\ =& 0.23315 \end{aligned}$$

🎯 **Result**

Part (a): $P(x) =_{12} C_x(0.52)^x(0.48)^{12-x}$

Part (b): $\mu = 6.24$ and $\sigma = 1.73$

Part (c): 0.1775

Part (d): 0.23315

PROBLEM STATEMENT

Chapter 6, Page 216

Step 1

(a) A hypergeometric distribution needs to be used when we are interested in the number of successes in a sample, when the sample comes from a population with S successes and $N - S$ failures.

In this case, we have a set of 25 returns of which 5 had charitable contributions and thus $25 - 5 = 20$ did not have charitable contributions. When we consider a success to be a charitable contribution, then the sample thus has indeed a **hypergeometric** distribution as we select a sample of 4 returns.

Step 2

(b) There are 5 charitable contributions, while we select 4 of the 25 returns.

$$\begin{aligned} N &= \text{Population size} = 25 \\ n &= \text{Number of draws} = 4 \\ S &= \text{Number of observed successes} = 5 \end{aligned}$$

Formula **hypergeometric probability**:

$$P(X = k) = \frac{SCX\,N-SCn-X}{NCn}$$

Since we need to select exactly 1 charitable contribution, we evaluate the definition of hypergeometric probability at $x = 1$:

$$\begin{aligned} P(X = 1) &= \frac{5C125-5C4-1}{25C4} \\ &= \frac{5C120C3}{25C4} \\ &= \frac{\frac{5!}{1!(5-1)!} \cdot \frac{20!}{3!(20-3)!}}{\frac{25!}{4!(25-4)!}} \\ &= \frac{5 \cdot 1140}{12,650} \\ &= \frac{5700}{12,650} \\ &= \frac{114}{253} \\ &\approx 0.4506 \\ &= 45.06\% \end{aligned}$$

Step 3

(c) There are 5 charitable contributions, while we select 4 of the 25 returns.

$$\begin{aligned} N &= \text{Population size} = 25 \\ n &= \text{Number of draws} = 4 \\ S &= \text{Number of observed successes} = 5 \end{aligned}$$

Formula **hypergeometric probability**:

$$P(X=k) = \frac{{}_{S}C_{X}\,{}_{N-S}C_{n-X}}{{}_{N}C_{n}}$$

We evaluate the definition of hypergeometric probability at $x=0$:

$$\begin{aligned} P(X=0) &= \frac{{}_{5}C_{0}\,{}_{25-5}C_{4-0}}{{}_{25}C_{4}} \\ &= \frac{{}_{5}C_{0}\,{}_{20}C_{4}}{{}_{25}C_{4}} \\ &= \frac{\frac{5!}{0!(5-0)!}\cdot\frac{20!}{4!(20-4)!}}{\frac{25!}{4!(25-4)!}} \\ &= \frac{1\cdot 4845}{12,650} \\ &= \frac{4845}{12,650} \\ &= \frac{969}{2530} \end{aligned}$$

Use the **Complement rule**: $P(\sim A) = P(\text{ not } A) = 1 - P(A)$

$$P(X \geq 1) = 1 - P(X=0) = 1 - \frac{969}{2530} = \frac{1561}{2530} \approx 0.6170 = 61.70\%$$

PROBLEM STATEMENT

Chapter 6, Page 216

Step 1

Total partners in the firm $N = 10$

Partners living in Northern Kentucky or number of successes (S)=3

Members of committee (n)=3

Number of successes in the committee= x

a. Probability that One member of the committee lives in northern Kentucky and the others live in Ohio i.e $P(x = 1)$

$$\begin{aligned} P(x=1) &= \frac{({}^{S}C_{x})({}^{N-S}C_{n-x})}{{}^{N}C_{n}} \\ &= \frac{({}^{3}C_{1})({}^{10-3}C_{3-1})}{{}^{10}C_{3}} \\ &= \frac{3 \times 21}{120} \\ &= 0.525 \end{aligned}$$

b. Probability that at least 1 member of the committee lives in northern Kentucky i.e $P(x \geq 1)$

$$\begin{aligned} P(x \geq 1) &= 1 - P(x < 1) \\ &= 1 - P(x = 0) \\ &= 1 - \frac{({}^{3}C_{0})({}^{10-3}C_{3-0})}{{}^{10}C_{3}} \\ &= 1 - \frac{1 \times 35}{120} \\ &= 1 - 0.29167 \\ &= 0.70833 \end{aligned}$$

Result

a. 0.525

b. 0.70833

PROBLEM STATEMENT

Chapter 6, Page 216

Step 1

Top 9 vehicles $N = 9$

Number of Honda vehicles in the top 9 or successes (S)=4

Sample of cars (n)=3

Number of successes in the sample = (x)

a. Probability distribution for the number of Hondas in a sample of three cars chosen from the top nine

Let $P(x)$ be the probability that x vehicles out of 3 are Honda

$$\begin{aligned} P(x) &= \frac{({}_S C_x)({}_{N-S} C_{n-x})}{{}_N C_n} \\ &= \frac{({}_4 C_x)({}_{9-4} C_{3-x})}{{}_9 C_3} \\ &= \frac{({}_4 C_x)({}_5 C_{3-x})}{84} \end{aligned}$$

b. Probability that at least one Honda is included i.e $P(x \geq 1)$

$$\begin{aligned} P(x \geq 1) &= 1 - P(x < 1) \\ &= 1 - P(x = 0) \\ &= 1 - \frac{({}_4 C_0)({}_5 C_{3-0})}{84} \\ &= 1 - \frac{1 \times 10}{84} \\ &= 1 - 0.12 \\ &= 0.88 \end{aligned}$$

Result

Part (a): $P(x) = \frac{({}_4 C_x)({}_5 C_{3-x})}{84}$

Part (b): 0.88

PROBLEM STATEMENT

Chapter 6, Page 216

Step 1

There are 4 females among 12 applicants, while we select 4 applicants.

$$\begin{aligned} N &= \text{Population size} = 12 \\ n &= \text{Number of draws} = 4 \\ S &= \text{Number of observed successes} = 4 \end{aligned}$$

Formula **hypergeometric probability**:

$$P(X=k) = \frac{SCXN-SCn-X}{NCn}$$

We evaluate the definition of hypergeometric probability at $x=0$:

$$\begin{aligned} P(X=0) &= \frac{4C012-4C4-0}{12C4} \\ &= \frac{4C08C4}{12C4} \\ &= \frac{\frac{4!}{0!(4-0)!}\cdot\frac{8!}{4!(8-4)!}}{\frac{12!}{4!(12-4)!}} \\ &= \frac{1\cdot 70}{495} \\ &= \frac{70}{495} \\ &= \frac{14}{99} \\ &\approx 0.1414 \\ &= 14.14\% \end{aligned}$$

We note that the probability is large (greater than 5%), which indicates that there was no discrimination.

Result

$$\frac{14}{99} \approx 0.1414 = 14.14\%$$

PROBLEM STATEMENT

Chapter 6, Page 216

Step 1

States with the largest population $N = 15$

Number of states having any coastline or number of successes (S)=10

Sample of states (n)=3

Number of successes in the sample =(x)

a. Probability that none of the states selected have any coastline i.e $(P(x = 0))$

$$\begin{aligned} P(x = 0) &= \frac{(SC_x)(N-SC_{n-x})}{NC_n} \\ &= \frac{(10C_0)(15-10C_3-0)}{15C_3} \\ &= \frac{1 \times 10}{455} \\ &= 0.022 \end{aligned}$$

b. Probability that exactly one of the selected states has a coastline i.e $(P(x = 1))$

$$\begin{aligned} P(x = 1) &= \frac{(SC_x)(N-SC_{n-x})}{NC_n} \\ &= \frac{(10C_1)(15-10C_3-1)}{15C_3} \\ &= \frac{10 \times 10}{455} \\ &= 0.22 \end{aligned}$$

c. Probability that at least one of the selected states has a coastline i.e $(P(x \geq 1))$

$$\begin{aligned} P(x \geq 1) &= 1 - P(x < 1) \\ &= 1 - P(x = 0) \\ &= 1 - 0.022 \text{ from } \mathbf{a.} \\ &= 0.978 \end{aligned}$$

Result

a. 0.022

b. 0.22

c. : 0.978

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

$$\mu = \text{mean} = 3 \text{ (per day)}$$

(a) Evaluate the probability of Poisson probability at $x = 0$:

$$\begin{aligned} P(X = 0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^0 e^{-3}}{0!} \\ &= \frac{1e^{-3}}{1} \\ &= e^{-3} \\ &\approx 0.0498 \\ &= 4.98\% \end{aligned}$$

Step 3

(b) Use the complement rule and the result of part (a):

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - 0.0498 \\ &= 0.9502 \end{aligned}$$

Assuming that the number of Lexus automobiles sold in a day is not affected by the sales on any other day, it is appropriate to used the multiplication rule for independent events:

$$\begin{aligned} &P(X \geq 1 \text{ on five consecutive days}) \\ &= P(X \geq 1) \times P(X \geq 1) \times P(X \geq 1) \times P(X \geq 1) \times P(X \geq 1) \\ &= (P(X \geq 1))^5 \\ &= 0.9502^5 \\ &\approx 0.7746 \\ &= 77.46\% \end{aligned}$$

🎯 **Result**

(a) $e^{-3} \approx 0.0498 = 4.98\%$

(b) $0.7746 = 77.46\%$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\begin{aligned}n&=\text{Sample size}=200\\ \pi&=\text{Probability of success}=1.5\%=0.015\end{aligned}$$

(a) The mean is the product of the sample size and the probability of success:

$$\mu=n\pi=200(0.015)=3$$

Evaluate the probability of Poisson probability at $x=0$:

$$\begin{aligned}P(X=0)&=\frac{\mu^k e^{-\mu}}{k!}\\ &=\frac{3^0 e^{-3}}{0!}\\ &=\frac{1e^{-3}}{1}\\ &=e^{-3}\\ &\approx 0.0498\\ &=4.98\%\end{aligned}$$

Step 3

(b) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 200(0.015) = 3$$

Evaluate the probability of Poisson probability at $x = 0, 1, 2$:

$$\begin{aligned} P(X=0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^0 e^{-3}}{0!} \\ &= \frac{1e^{-3}}{1} \\ &= e^{-3} \\ &\approx 0.0498 \\ P(X=1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^1 e^{-3}}{1!} \\ &= \frac{3e^{-3}}{1} \\ &= 3e^{-3} \\ &\approx 0.1494 \\ P(X=2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^2 e^{-3}}{2!} \\ &= \frac{9e^{-3}}{2} \\ &= \frac{9}{2}e^{-3} \\ &\approx 0.2240 \end{aligned}$$

Step 4

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same set of 200 antennas):

$$\begin{aligned} P(X<3) &= P(X=0) + P(X=1) + P(X=2) \\ &= 0.0498 + 0.1494 + 0.2240 \\ &= 0.4232 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 3) &= 1 - P(X<3) \\ &= 1 - 0.4232 \\ &= 0.5768 \\ &= 57.68\% \end{aligned}$$

Result

(a) $e^{-3} \approx 0.0498 = 4.98\%$

(b) $0.5768 = 57.68\%$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\mu=\text{Mean}=4$$

(a) Evaluate the probability of Poisson probability at $x=0$:

$$\begin{aligned}P(X=0)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{4^0e^{-4}}{0!}\\&=\frac{1e^{-4}}{1}\\&=e^{-4}\\&\approx 0.0183\\&=1.83\%\end{aligned}$$

Step 3

(b) Evaluate the probability of Poisson probability at $x=4$:

$$\begin{aligned}P(X=4)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{4^4e^{-4}}{4!}\\&=\frac{256e^{-4}}{24}\\&=\frac{32}{3}e^{-4}\\&\approx 0.1954\\&=19.54\%\end{aligned}$$

Step 4

(c) Evaluate the probability of Poisson probability at $x = 1, 2, 3$:

$$\begin{aligned} P(X=1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{4^1 e^{-4}}{1!} \\ &= \frac{4e^{-4}}{1} \\ &= 4e^{-4} \\ &\approx 0.0733 \\ P(X=2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{4^2 e^{-4}}{2!} \\ &= \frac{16e^{-4}}{2} \\ &= 8e^{-4} \\ &\approx 0.1465 \\ P(X=3) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{4^3 e^{-4}}{3!} \\ &= \frac{64e^{-4}}{6} \\ &= \frac{32}{3}e^{-4} \\ &\approx 0.1954 \end{aligned}$$

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same day) and use the results of part (a)-(b):

$$\begin{aligned} P(X \le 4) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &= 0.0183 + 0.0733 + 0.1465 + 0.1954 + 0.1954 \\ &= 0.6289 \\ &= 62.89\% \end{aligned}$$

Step 5

(d) Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same day) and use the results of part (a)-(c):

$$\begin{aligned} P(X \le 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= 0.0183 + 0.0733 + 0.1465 + 0.1954 \\ &= 0.4335 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \ge 4) &= 1 - P(X \le 3) \\ &= 1 - 0.4335 \\ &= 0.5665 \\ &= 56.65\% \end{aligned}$$

Result

(a) $e^{-4} \approx 0.0183 = 1.83\%$

(b) $\frac{32}{3}e^{-4} \approx 0.1954 = 19.54\%$

(c) $0.5665 = 56.65\%$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\mu=\text{Mean}=2$$

(a) Evaluate the probability of Poisson probability at $x=1$:

$$\begin{aligned}P(X=1)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{2^1 e^{-2}}{1!}\\&=\frac{2e^{-2}}{1}\\&=2e^{-2}\\&\approx 0.2707\\&=27.07\%\end{aligned}$$

Step 3

(b) Evaluate the probability of Poisson probability at $x = 0, 2, 3, 4$:

$$\begin{aligned}
P(X=0) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{2^0 e^{-2}}{0!} \\
&= \frac{1e^{-2}}{1} \\
&= e^{-2} \\
&\approx 0.1353 \\
P(X=2) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{2^2 e^{-2}}{2!} \\
&= \frac{4e^{-2}}{2} \\
&= 2e^{-2} \\
&\approx 0.2707 \\
P(X=3) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{2^3 e^{-2}}{3!} \\
&= \frac{8e^{-2}}{6} \\
&= \frac{4}{3}e^{-2} \\
&\approx 0.1804 \\
P(X=4) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{2^4 e^{-2}}{4!} \\
&= \frac{16e^{-2}}{24} \\
&= \frac{2}{3}e^{-2} \\
&\approx 0.0902
\end{aligned}$$

Step 4

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same hour) and the result of part (a):

$$\begin{aligned}
P(X<5) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) \\
&= 0.1353 + 0.2707 + 0.2707 + 0.1804 + 0.0902 \\
&= 0.9473
\end{aligned}$$

Use the complement rule:

$$\begin{aligned}
P(X \geq 5) &= 1 - P(X<5) \\
&= 1 - 0.9473 \\
&= 0.0527 \\
&= 5.27\%
\end{aligned}$$

Step 5

(c) Evaluate the probability of Poisson probability at $x = 0, 2, 3, 4$:

$$\begin{aligned}
P(X=0) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{2^0 e^{-2}}{0!} \\
&= \frac{1e^{-2}}{1} \\
&= e^{-2} \\
&\approx 0.1353 \\
&= 13.53\%
\end{aligned}$$

🎯 **Result**

(a) $2e^{-2} \approx 0.2707 = 27.07\%$

(b) $0.0527 = 5.27\%$

(c) $e^{-2} \approx 0.1353 = 13.53\%$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\mu=\text{Mean}=3.1$$

(a) Evaluate the probability of Poisson probability at $x=4$:

$$\begin{aligned}P(X=4)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{3.1^4 e^{-3.1}}{4!}\\&=\frac{92.3521e^{-3.1}}{24}\\&\approx 0.1733\\&=17.33\%\end{aligned}$$

Step 3

(b) Evaluate the probability of Poisson probability at $x=0$:

$$\begin{aligned}P(X=0)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{3.1^0 e^{-3.1}}{0!}\\&=\frac{1e^{-3.1}}{1}\\&=e^{-3.1}\\&\approx 0.0450\\&=4.50\%\end{aligned}$$

Step 4

(c) Use the complement rule :

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - e^{-3.1} \\ &= 0.9550 \\ &= 95.50\% \end{aligned}$$

Result

(a) $0.1733 = 17.33\%$

(b) $e^{-3.1} \approx 0.0450 = 4.50\%$

(c) $1 - e^{-3.1} = 0.9550 = 95.50\%$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

$$\mu=\text{Mean}=2$$

(a) Evaluate the probability of Poisson probability at $x=0,1,2,3,4$:

$$\begin{aligned}
P(X=0)&=\frac{\mu^k e^{-\mu}}{k!}\\
&=\frac{2^0 e^{-2}}{0!}\\
&=\frac{2e^{-2}}{1}\\
&=e^{-2}\\
&\approx 0.1353\\
P(X=1)&=\frac{\mu^k e^{-\mu}}{k!}\\
&=\frac{2^1 e^{-2}}{1!}\\
&=\frac{2e^{-2}}{1}\\
&=2e^{-2}\\
&\approx 0.2707\\
P(X=2)&=\frac{\mu^k e^{-\mu}}{k!}\\
&=\frac{2^2 e^{-2}}{2!}\\
&=\frac{4e^{-2}}{2}\\
&=2e^{-2}\\
&\approx 0.2707
\end{aligned}$$

Step 3

$$\begin{aligned}
P(X=3) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{2^3 e^{-2}}{3!} \\
&= \frac{8e^{-2}}{6} \\
&= \frac{4}{3}e^{-2} \\
&\approx 0.1804 \\
P(X=4) &= \frac{\mu^k e^{-\mu}}{k!} \\
&= \frac{2^4 e^{-2}}{4!} \\
&= \frac{16e^{-2}}{24} \\
&= \frac{2}{3}e^{-2} \\
&\approx 0.0902
\end{aligned}$$

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same month):

$$\begin{aligned}
P(X<5) &= P(X=0) + P(X=1) + P(X=2) \\
&= P(X=3) + P(X=4) \\
&= 0.1353 + 0.2707 + 0.2707 \\
&+ 0.1804 + 0.0902 \\
&= 0.9473 \\
&= 94.73\%
\end{aligned}$$

We note that the probability of fever than 5 unfilled order is slightly less than 95% and thus New Process Inc. did not live up to its internal goal (although they only just did not making the goal).

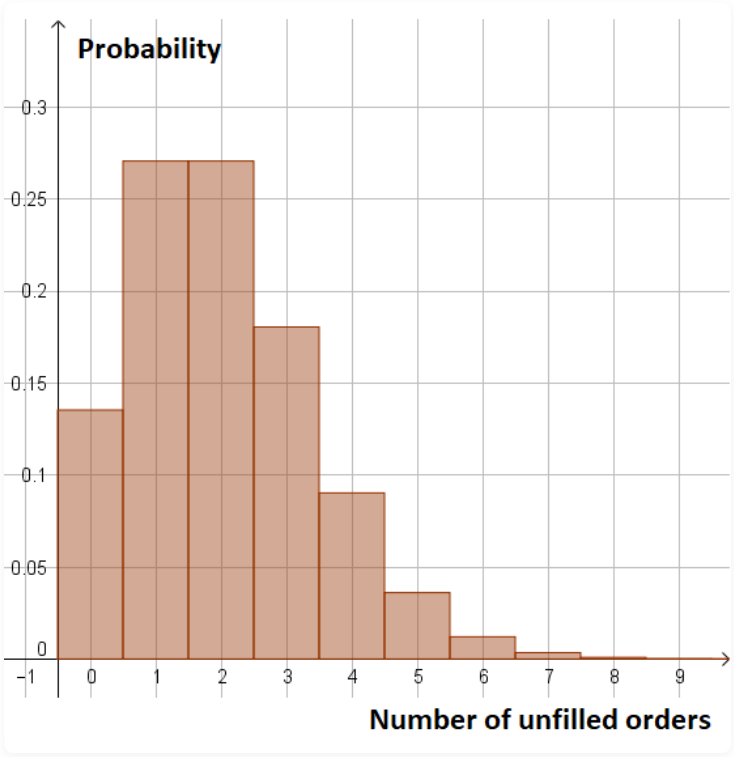
Step 4

(b)

Histogram

The number of unfilled orders is on the horizontal axis and the probability is on the vertical axis.

All bars need to have the same width, while the bars need to be centered at the number of unfilled orders.



Result

(a) No

(b) Histogram

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

2 of the first 113 missions were a failure.

$$n=\text{Sample size}=23$$
$$\pi=\text{Probability of success}=\frac{2}{113}$$

The mean is the product of the sample size and the probability of success:

$$\mu=n\pi=23\times\frac{2}{113}\approx 0.4071$$

Evaluate the probability of Poisson probability at $x=0,2$:

$$\begin{aligned}P(X=2)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{0.4071^2 e^{-0.4071}}{2!}\\&=\frac{0.4071^2 e^{-0.4071}}{2}\\&\approx 0.0552\\&=5.52\%\end{aligned}$$

$$\begin{aligned}P(X=0)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{0.4071^0 e^{-0.4071}}{0!}\\&=\frac{1e^{-0.4071}}{1}\\&=e^{-0.4071}\\&\approx 0.6656\\&=66.56\%\end{aligned}$$

Result

2 failures: $0.0552=5.52\%$

0 failures: $0.6656=66.56\%$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Step 2

SOLUTION

$$n=\text{Sample size}=34$$
$$\pi=\text{Probability of success}=0.50$$

The mean is the product of the sample size and the probability of success:

$$\mu=n\pi=34(0.50)=17$$

Evaluate the probability of Poisson probability at $x=29$:

$$\begin{aligned}P(X=29)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{17^{29}e^{-17}}{29!}\\&\approx 0.0023\\&=0.23\%\end{aligned}$$

Result

$$0.0023=0.23\%$$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 155$$
$$\pi = \text{Probability of success} = \frac{1}{3709}$$

The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 155 \times \frac{1}{3709} \approx 0.0418$$

Evaluate the probability of Poisson probability at $x = 4$:

$$\begin{aligned} P(X = 4) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{0.0418^4 e^{-0.0418}}{4!} \\ &= \frac{0.0418^4 e^{-0.0418}}{24} \\ &\approx 0.000000122 \\ &= 0.0000122\% \end{aligned}$$

Note: The solution in the back of the book is incorrect, because it calculates $P(5)$ instead of $P(4)$.

Result

$$0.000000122 = 0.0000122\%$$

PROBLEM STATEMENT

Chapter 6, Page 217

Step 1

DEFINITIONS

Definition **binomial probability**:

$$P(X=k) = {}_n C_k \cdot \pi^k \cdot (1-\pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1-\pi)^{n-k}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

(a) The number of successes among a fixed number of independent trials with a constant probability of success follows a **binomial distribution**.

$$p = \text{Probability of successes} = 0.95$$

Assuming that we have a certain number of hurricanes among which we want to determine the number of successes, it is appropriate to use the binomial distribution.

Note: In part (b) and part (c), we note that the fixed number of trials (hurricanes) is 10.

Step 3

(b) Given:

$$\begin{aligned} n &= \text{Sample size} = 10 \\ \pi &= \text{Probability of success} = 0.95 \end{aligned}$$

Evaluate the definition of binomial probability at $k = 10$:

$$\begin{aligned} P(X=10) &= {}_{10} C_{10} \cdot 0.95^{10} \cdot (1-0.95)^{10-10} \\ &= \frac{10!}{10!(10-10)!} \cdot 0.95^{10} \cdot 0.05^{0} \\ &= 1 \cdot 0.95^{10} \cdot 1 \\ &= 0.95^{10} \\ &\approx 0.5987 \\ &= 59.87\% \end{aligned}$$

Step 4

(c) Given:

$$n = \text{Sample size} = 10$$
$$\pi = \text{Probability of success} = 0.95$$

Evaluate the definition of binomial probability at $k = 0$:

$$\begin{aligned} P(X = 0) &= {}_{10}C_0 \cdot 0.95^0 \cdot (1 - 0.95)^{10-0} \\ &= \frac{10!}{0!(10-0)!} \cdot 0.95^0 \cdot 0.05^{10} \\ &= 1 \cdot 1 \cdot 0.05^{10} \\ &= 0.05^{10} \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - 0.05^{10} \\ &\approx 1 \\ &= 100\% \end{aligned}$$

Result

(a) Binomial distribution

(b) $0.5987 = 59.87\%$

(c) Approximately 1 or 100%

PROBLEM STATEMENT

Chapter 6, Page 218

Step 1

DEFINITIONS

Definition **binomial probability**:

$$P(X=k) = {}_n C_k \cdot \pi^k \cdot (1-\pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1-\pi)^{n-k}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 15$$
$$\pi = \text{Probability of success} = 67\% = 0.67$$

(a) The mean (expected value) of a binomial distribution is the product of the sample size n and the probability π:

$$\mu = n\pi = 15 \times 0.67 = 10.05$$

The standard deviation of a binomial distribution is the square root of the product of the sample size n and the probabilities π and $1-\pi$ (that is, the standard deviation is the square root of the variance).

$$\sigma = \sqrt{n\pi(1-\pi)} = \sqrt{15(0.67)(1-0.67)} \approx 1.8211$$

Thus we expect 10.05 pennies, with a standard deviation of 1.8211 pennies.

Step 3

(b) Evaluate the definition of binomial probability at $k = 8$:

$$\begin{aligned} P(X=8) &= {}_{15} C_8 \cdot 0.67^8 \cdot (1-0.67)^{15-8} \\ &= \frac{15!}{8!(15-8)!} \cdot 0.67^8 \cdot 0.33^7 \\ &= 6435 \cdot 0.67^8 \cdot 0.33^7 \\ &\approx 0.1114 \\ &= 11.14\% \end{aligned}$$

Step 4

(c) Evaluate the definition of binomial probability at $k = 9, 10, 11, 12, 13, 14, 15$:

$$\begin{aligned}
P(X=9) &= {}_{15}C_9 \cdot 0.67^9 \cdot (1-0.67)^{15-9} \\
&= \frac{15!}{9!(15-9)!} \cdot 0.67^9 \cdot 0.33^6 \\
&= 5005 \cdot 0.67^9 \cdot 0.33^6 \\
&\approx 0.1759 \\
P(X=10) &= {}_{15}C_{10} \cdot 0.67^{10} \cdot (1-0.67)^{15-10} \\
&= \frac{15!}{10!(15-10)!} \cdot 0.67^{10} \cdot 0.33^5 \\
&= 3003 \cdot 0.67^{10} \cdot 0.33^5 \\
&\approx 0.2142 \\
P(X=11) &= {}_{15}C_{11} \cdot 0.67^{11} \cdot (1-0.67)^{15-11} \\
&= \frac{15!}{11!(15-11)!} \cdot 0.67^{11} \cdot 0.33^4 \\
&= 1365 \cdot 0.67^{11} \cdot 0.33^4 \\
&\approx 0.1977 \\
P(X=12) &= {}_{15}C_{12} \cdot 0.67^{12} \cdot (1-0.67)^{15-12} \\
&= \frac{15!}{12!(15-12)!} \cdot 0.67^{12} \cdot 0.33^3 \\
&= 455 \cdot 0.67^{12} \cdot 0.33^3 \\
&\approx 0.1338 \\
P(X=13) &= {}_{15}C_{13} \cdot 0.67^{13} \cdot (1-0.67)^{15-13} \\
&= \frac{15!}{13!(15-13)!} \cdot 0.67^{13} \cdot 0.33^2 \\
&= 105 \cdot 0.67^{13} \cdot 0.33^2 \\
&\approx 0.0627
\end{aligned}$$

Step 5

$$\begin{aligned}
P(X=14) &= {}_{15}C_{14} \cdot 0.67^{14} \cdot (1-0.67)^{15-14} \\
&= \frac{15!}{14!(15-14)!} \cdot 0.67^{14} \cdot 0.33^1 \\
&= 15 \cdot 0.67^{14} \cdot 0.33 \\
&\approx 0.0182 \\
P(X=15) &= {}_{15}C_{15} \cdot 0.67^{15} \cdot (1-0.67)^{15-15} \\
&= \frac{15!}{15!(15-15)!} \cdot 0.67^{15} \cdot 0.33^0 \\
&= 1 \cdot 0.67^{15} \cdot 1 \\
&\approx 0.0025
\end{aligned}$$

Use the addition rule for mutually exclusive events (which is appropriate as you can't obtain two different number of successes for the same sample).

$$\begin{aligned}
P(X \geq 8) &= P(X=8) + P(X=9) + P(X=10) + P(X=11) \\
&\quad + P(X=12) + P(X=13) + P(X=14) + P(X=15) \\
&= 0.1114 + 0.1759 + 0.2142 + 0.1977 \\
&\quad + 0.1338 + 0.0627 + 0.0182 + 0.0025 \\
&= 0.9163 \\
&= 91.63\%
\end{aligned}$$

Result

(a) 10.05 pennies, 1.8211 pennies

(b) $0.1114 = 11.14\%$

(c) $0.9163 = 91.63\%$

PROBLEM STATEMENT

Chapter 6, Page 218

Step 1

Given:

$$n = \text{Sample size} = 105$$

(a) The frequency is the number of times that the data value occurs in the given data set.

The probability is the frequency divided by the sample size n.

x	Frequency	Probability $P(X = x)$
2	24	$\frac{24}{105}$
3	26	$\frac{26}{105}$
4	26	$\frac{26}{105}$
5	11	$\frac{11}{105}$
6	14	$\frac{14}{105}$
7	2	$\frac{2}{105}$
8	2	$\frac{2}{105}$

The expected value (or mean) is the sum of the product of each possibility x with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x)\\ &= 2 \times \frac{24}{105} + 3 \times \frac{26}{105} + 4 \times \frac{26}{105} + 5 \times \frac{11}{105}\\ &\quad + 6 \times \frac{14}{105} + 7 \times \frac{2}{105} + 8 \times \frac{2}{105}\\ &= \frac{399}{105}\\ &= \frac{19}{5}\\ &= 3.8\end{aligned}$$

Step 2

The variance is the expected value of the squared deviation from the mean:

$$\begin{aligned}\sigma^2 &= \sum (x-\mu)^2 P(x) \\ &= (2-3.8)^2 \times \frac{24}{105} + (3-3.8)^2 \times \frac{26}{105} + (4-3.8)^2 \times \frac{26}{105} + (5-3.8)^2 \times \frac{11}{105} \\ &\quad + (6-3.8)^2 \times \frac{14}{105} + (7-3.8)^2 \times \frac{2}{105} + (8-3.8)^2 \times \frac{2}{105} \\ &= \frac{1174}{525} \\ &\approx 2.2362\end{aligned}$$

The standard deviation is the square root of the variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{2.2362} \approx 1.4954$$

Step 3

(b) The frequency is the number of times that the data value occurs in the given data set.

The probability is the frequency divided by the sample size n.

x	Frequency	Probability $P(X = x)$
1.5	16	$\frac{16}{105}$
2	65	$\frac{65}{105}$
2.5	15	$\frac{15}{105}$
3	9	$\frac{9}{105}$

The expected value (or mean) is the sum of the product of each possibility x with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x) \\ &= 1.5 \times \frac{16}{105} + 2 \times \frac{65}{105} + 2.5 \times \frac{15}{105} + 3 \times \frac{9}{105} \\ &= \frac{218.5}{105} \\ &= \frac{437}{210} \\ &\approx 2.0810\end{aligned}$$

Step 4

The variance is the expected value of the squared deviation from the mean:

$$\begin{aligned}\sigma^2 &= \sum (x-\mu)^2 P(x) \\ &= (1.5-2.0810)^2 \times \frac{16}{105} + (2-2.0810)^2 \times \frac{65}{105} \\ &+ (2.5-2.0810)^2 \times \frac{15}{105} + (3-2.0810)^2 \times \frac{9}{105} \\ &\approx 0.1530\end{aligned}$$

The standard deviation is the square root of the variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{0.1530} \approx 0.3912$$

Result

(a) Mean 3.8 and Standard deviation 1.4954

(b) Mean 2.0810 and Standard deviation 0.3912

PROBLEM STATEMENT

Chapter 6, Page 218

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A)=P(\text{not } A)=1-P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A\cup B)=P(A \text{ or } B)=P(A)+P(B)$$

Step 2

SOLUTION

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned}\overline{x}&=\frac{\sum_{i=1}^{n}x_i}{n}\\&=\frac{212+161+145+...+172}{30}\\&=\frac{5042}{30}\\&\approx 168.0667\end{aligned}$$

We have been given that the mean of the Poisson distribution will be the sample mean divided by 162 and multiplied by 2.

$$\mu=\text{Mean}\times\frac{2}{162}=168.0667\times\frac{1}{81}\approx 2.0749$$

(a) Evaluate the probability of Poisson probability at $x=0$:

$$\begin{aligned}P(X=0)&=\frac{\mu^k e^{-\mu}}{k!}\\&=\frac{2.0749^0 e^{-2.0749}}{0!}\\&=\frac{1e^{-2.0749}}{1}\\&=e^{-2.0749}\\&\approx 0.1256\\&=12.56\%\end{aligned}$$

Step 3

(b) Evaluate the probability of Poisson probability at $x = 2$:

$$\begin{aligned} P(X = 2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2.0749^2 e^{-2.0749}}{2!} \\ &= \frac{2.0749^2 e^{-2.0749}}{2} \\ &\approx 0.2703 \\ &= 27.03\% \end{aligned}$$

Step 4

(c) Evaluate the probability of Poisson probability at $x = 1, 3$:

$$\begin{aligned} P(X = 1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2.0749^1 e^{-2.0749}}{1!} \\ &= \frac{2.0749 e^{-2.0749}}{1} \\ &= 2.0749 e^{-2.0749} \\ &\approx 0.2605 \\ P(X = 3) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2.0749^3 e^{-2.0749}}{3!} \\ &= \frac{2.0749^3 e^{-2.0749}}{6} \\ &\approx 0.1869 \end{aligned}$$

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same sample) and use the results of part (a)-(b):

$$\begin{aligned} P(X < 4) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ &= 0.1256 + 0.2605 + 0.2703 + 0.1869 \\ &= 0.8433 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 4) &= 1 - P(X < 4) \\ &= 1 - 0.8433 \\ &= 0.1567 \\ &= 15.67\% \end{aligned}$$

Result

(a) $e^{-2.0749} \approx 0.1256 = 12.56\%$

(b) $0.2703 = 27.03\%$

(c) $0.1567 = 15.67\%$